Cutting the Flow: Geopolitical Disruptions to Russian Oil Supply and the Tails of Euro Area Inflation

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Abstract

Geopolitical disruptions to Russian oil supply do not move euro area inflation uniformly: they reshape its distribution. Using an AI-based news index of oil supply disruptions and a quantile factor-augmented VAR for the 21 member states, I show that a disruption first stretches the upper tail of inflation, raising upside risk on impact, before the median and the lower tail catch up. Responses differ widely across member states and line up with pre-war reliance on Russian energy. The disruptions of 2022–2023 added about two percentage points to euro area inflation at their peak. Size matters: before the invasion, disruptions were disinflationary, and only disruptions of the scale seen in 2022 have effects on inflation that are far more than proportional to their size.

Keywords: geopolitical risk, oil supply disruptions, inflation risk, quantile FAVAR, euro area
JEL codes: C32, E31, E52, F51, Q43

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1 Introduction

> *"The war in Ukraine is severely affecting the euro area economy and has significantly increased uncertainty. [. . .] The upside risks surrounding the inflation outlook have also intensified, especially in the near term."*
>
> ***— Christine Lagarde, ECB Press Conference, April 2022***

Russia's invasion of Ukraine on 24 February 2022 set off a sequence of geopolitical events that disrupted Europe's energy supply (Figure 1). Within months, the European Union agreed a phased embargo on Russian oil and launched REPowerEU to end its dependence on Russian energy, while Russia curtailed pipeline gas deliveries and the Nord Stream pipelines were sabotaged. Following the European Union embargoes on crude oil and refined products and the G7 price cap, euro area imports of Russian oil fell by around 90% between 2021 and 2023.[1]

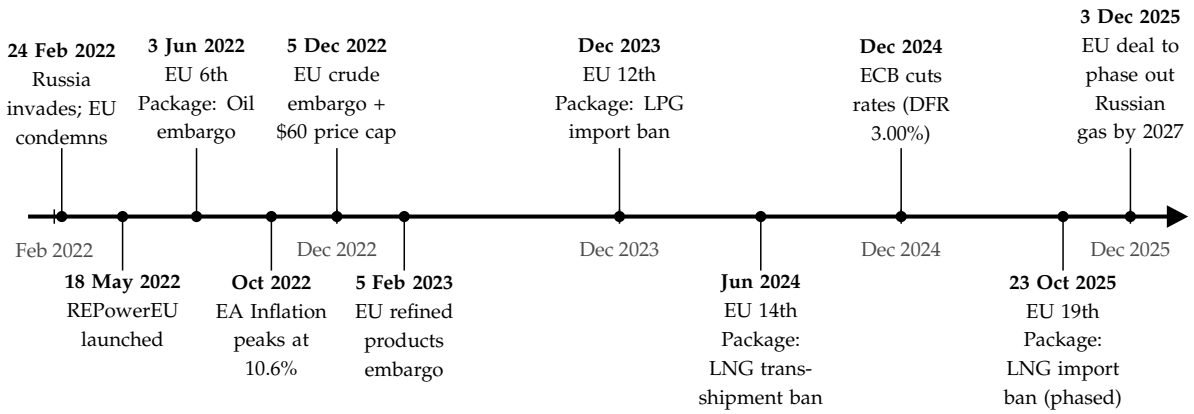


Figure 1: Selected EU Sanctions, Energy Phase-Out, and Euro Area Economic Milestones

Note: The figure tracks key European Union (EU) and euro area (EA) policy and macroeconomic milestones following Russia's invasion of Ukraine.
Sources: European Council; European Commission; European Central Bank (ECB); Eurostat.

Over the same period, inflation in the euro area rose to rates not seen since the creation of the monetary union, peaking at 10.6% in October 2022. Policymakers and analysts attributed a large part of the surge to the disruption of energy supply that fol-

[1] According to Eurostat, imports of oil and petroleum products from Russia fell by 87% in the European Union and by about 93% in the euro area between 2021 and 2023 (Table B2 in the Appendix).

lowed the invasion.[2] For a central bank, however, the average inflation rate is not the whole of the problem. Geopolitical risk not only raises average inflation but also increases inflation uncertainty and the likelihood of large inflation surges (Caldara et al., 2025). A disruption of the kind observed in 2022 need not shift the inflation distribution uniformly. It may instead stretch its upper tail, making high-inflation outcomes more likely. Nor need the member states respond alike. As Figure 6 shows, their reliance on Russian oil and natural gas differed widely before the war, so a common disruption may reshape inflation risk differently across the euro area.

Main findings. A geopolitical disruption to Russian oil supply raises inflation across the whole distribution, but not uniformly. The upper tail moves first. On impact, the 90th percentile of euro area inflation rises while the median, if anything, falls, so that upside risk increases sharply. Within a few months, the median and the 10th percentile catch up, and all three percentiles peak after about half a year before fading after around 14 months. The increase in upside risk is therefore short-lived, and over longer horizons the shock shifts the distribution rather than changing its shape. The responses of the global factors indicate that both a supply and a demand channel are at work. Gas prices rise sharply and monetary policy tightens, consistent with the supply channel, while financial stress rises, supply chain pressures ease, and unemployment increases gradually, consistent with weaker demand. The response is far from uniform across member states. In Germany, Belgium, and other member states, the upper tail jumps on impact and upside risk rises briefly; in the Baltic states and other eastern members, the response builds up more gradually but extends to the whole distribution and reaches larger peaks. These differences are associated with pre-war reliance on Russian oil and gas: more exposed member states experience larger responses, especially in the lower part of the distribution. A counterfactual exercise indicates that the disruptions of 2022 and 2023 added about two percentage points to euro area inflation at its peak, in mid-2022, and about three quarters of a percentage point on average over the two years, with the largest contributions in the Baltic states, among the member states most exposed to Russian energy. Finally, the inflationary effect is specific to large disruptions. In samples that end before the invasion, disruptions are disinflationary, and nonlinear estimates show that only disruptions of the size observed in 2022 raise oil prices and supply chain pressures, with effects on inflation and on its dispersion that are far more than proportional to their size.

[2] See Arce et al. (2023) for a discussion of the contribution of energy prices to the 2022 inflation surge.

These findings suggest that supply disruptions should be assessed through the risks around the inflation outlook and not only through its central path, and that reducing dependency on a single energy supplier lowers the exposure of member states to such disruptions.

Connections to the literature. The paper contributes to four strands of the literature. First, it contributes to the literature on the macroeconomic effects of geopolitical risk. Building on the news-based index of Caldara and Iacoviello (2022), a growing body of work documents that geopolitical risk shocks depress economic activity, raise uncertainty, and transmit to oil prices and inflation (Caldara and Iacoviello, 2022; Caldara et al., 2025). A long tradition identifies exogenous oil supply shocks from wars and political events (Hamilton, 2003; Kilian, 2008), and a recent literature focuses specifically on geopolitical oil supply shocks. Verduzco-Bustos and Zanetti (2026) show that they generate much larger oil price increases per unit of lost production than conventional supply shocks and spill over most strongly to oil-importing economies that depend heavily on foreign energy, while Pinchetti (2025) shows that geopolitical shocks transmitted through energy markets are inflationary, whereas those that are not are deflationary. These studies focus on average outcomes, mostly in the United States. I narrow the focus to geopolitical disruptions to Russian oil supply, measured with the AI-based index of Iacoviello and Tong (2026), and study how they affect the entire distribution of inflation, in the Euro Area and in each of its member states.

Second, the paper contributes to the literature on the heterogeneous transmission of common shocks within the Euro Area. A large body of work documents that a single monetary policy is transmitted unevenly across member states, owing to differences in financial structure, labor markets, and the composition of consumption (Ciccarelli et al., 2013; Burriel and Galesi, 2018; Corsetti et al., 2022). Closest to my approach, Boivin et al. (2009), Barigozzi et al. (2014), and Bénassy-Quéré et al. (2026) use factor models to recover the responses of individual member states to a common monetary policy shock, building on the evidence that national inflation rates share a strong common component (Ciccarelli and Mojon, 2010). Cross-country differences in the effects of oil shocks have also been documented outside the factor framework (Peersman and Van Robays, 2012). I show that a common geopolitical supply disruption reshapes the inflation distribution differently across member states, in both the timing and the size of the response, and that these differences are associated with pre-war reliance on Russian oil and gas. Because the quantile factor-augmented VAR

recovers the response of each member state's inflation quantiles to a single identified shock, the heterogeneity concerns not only expected inflation but also inflation risk.

Third, the paper contributes to the literature on how shocks shift the tails of macroeconomic distributions. Following the growth-at-risk framework of Adrian et al. (2019), a large literature characterizes how conditions and shocks move the tails of the future output distribution rather than its mean. Forni et al. (2024b) combine quantile regressions with structural VARs to trace identified shocks through the conditional distribution of growth, decomposing uncertainty into upside and downside components defined, as I define inflation risk, by the gaps between the median and the tail quantiles; related work extends this to monetary policy (Forni et al., 2024a). On the inflation side, López-Salido and Loria (2024) bring the tail-risk lens to the conditional inflation distribution, finding that financial conditions carry substantial inflation risk in the United States and the Euro Area. Using the quantile factor-augmented VAR of Korobilis and Schröder (2025), I contribute by tracing a geopolitical, supply-side shock through the inflation distribution of the Euro Area and of each member state. The shock widens the upper tail on impact, before the median and the lower tail follow, and the responses of energy prices, financial stress, and unemployment indicate that both a supply and a demand channel are at work.

Finally, the paper relates to the literature on the role of shock size. A recurring theme in the geopolitical risk literature is that the consequences of a shock depend on its scale: minor, localized events have limited macroeconomic repercussions, whereas large-scale shocks propagate disproportionately through the economy. Brignone et al. (2026) formalize this "size matters" logic and show that the macroeconomic effects of geopolitical risk are strongly nonlinear in the magnitude of the shock. Following their approach, I show that the same logic governs geopolitical disruptions to Russian oil supply and helps explain why the 2022 disruption was inflationary while earlier ones were not. Only disruptions well above the size of the pre-war episodes raise oil prices and supply chain pressures, and disruptions of the size observed in 2022 have effects on inflation and on its dispersion that are far more than proportional to their size.

Outline of the paper. The rest of the paper is organized as follows. Section 2 presents the quantile factor-augmented VAR, the data, and the identification strategy. Section 3 reports the main results for the euro area and its member states and relates the differences across member states to their exposure to Russian energy. Section 4 examines the role of the 2022 disruption, its contribution to the inflation surge, and the size of

the shock. Section 5 concludes.

2 The Model

Let π_{it} denote headline inflation rate in euro area member state $i = 1, \ldots, n$ in month $t = 1, \ldots, T$, with $n = 21$. Unless otherwise stated, all series are standardized before estimation by subtracting their mean and dividing by their standard deviation, country by country for the inflation panel. I characterize the distribution of the $n \times 1$ vector $\boldsymbol{\pi}_t = [\pi_{1t}, \ldots, \pi_{nt}]'$ through a latent inflation factor $\Pi_{t,(q)}$ specific to each quantile level q, which summarizes the common dynamics of the panel at that point of the distribution, and a $k \times 1$ vector of observed global factors $\boldsymbol{\mathcal{G}}_t$. The q-th conditional quantile of π_{it} is linear in both,

$$\mathcal{Q}_q \left[\pi_{it} \mid \Pi_{t,(q)}, \boldsymbol{\mathcal{G}}_t \right] = \gamma_{i(q)} \Pi_{t,(q)} + \boldsymbol{\psi}_{i(q)} \boldsymbol{\mathcal{G}}_t \tag{1}$$

where $\gamma_{i(q)}$ is the loading of country i on the quantile-specific latent inflation factor and $\boldsymbol{\psi}_{i(q)}$ is a $1 \times k$ vector of loadings on the observed global factors.

The vector of global factors is defined as

$$\underbrace{\boldsymbol{\mathcal{G}}_t}_{7\times 1} = \left[\mathcal{S}_t \quad \pi_t^{oil} \quad \pi_t^{gas} \quad \mathcal{F}_t \quad gscpi_t \quad u_t \quad R_t \right]'$$

where $\mathcal{S}_t$ is the Russia component of the geopolitical oil supply disruption subindex of Iacoviello and Tong (2026), which measures the share of newspaper articles discussing disruptions to oil and energy supply from Russia caused by geopolitical events, with each article weighted by its geopolitical risk score[3], π_t^{oil} and π_t^{gas} are the year-on-year growth rates of the Brent crude oil price and of the European natural gas price, both in euros, $\mathcal{F}_t$ is the euro area composite indicator of systemic stress (Holló et al., 2012; Chavleishvili and Kremer, 2023), $gscpi_t$ is the global supply chain pressure index

[3]The index extends the news-based geopolitical risk index of Caldara and Iacoviello (2022). Rather than counting articles that contain geopolitical keywords, it uses a large language model (GPT-4o mini) to assign a geopolitical risk score to about 4.6 million articles published in three major US newspapers since 1960, which reduces false positives from articles that mention war or terrorism in a non-geopolitical context. The oil supply disruption subindex is built in two steps. Among the articles with a geopolitical risk score above 0.5 that mention oil- or energy-related keywords, a second classification retains those discussing an oil or energy supply disruption caused by a geopolitical event and identifies the regions affected. The Russia component is based on the articles in which Russia is among the affected regions. Figure C1 plots the index.

of the Federal Reserve Bank of New York (Benigno et al., 2022), u_t is the euro area unemployment rate, and R_t is the euro area shadow rate of Wu and Xia (2020), which measures the monetary policy stance and remains informative when policy rates are constrained by the lower bound. Outside the period in which the shadow rate is available, R_t is the overnight rate: EONIA before September 2004 and the euro short-term rate (€STR) after December 2022.[4]

Following the literature on macroeconomic risk, I focus on three quantile levels that characterize the left tail, the center, and the right tail of the conditional inflation distribution. Collecting the three latent factors in $\boldsymbol{\Pi}_t = [\Pi_{t,(0.1)}, \Pi_{t,(0.5)}, \Pi_{t,(0.9)}]'$, the global and and quantile factors evolve jointly according to a VAR(p),

$$\boldsymbol{y}_t \equiv \begin{bmatrix} \boldsymbol{\mathcal{G}}_t \\ \boldsymbol{\Pi}_t \end{bmatrix} = \sum_{j=1}^{p} \boldsymbol{A}_j \boldsymbol{y}_{t-j} + \boldsymbol{e}_t, \quad \boldsymbol{e}_t = \boldsymbol{B}_0 \boldsymbol{\varepsilon}_t \tag{2}$$

where $\boldsymbol{A}_j$ are 10×10 matrices of autoregressive coefficients, $\boldsymbol{e}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_e)$ are the reduced-form innovations with $\boldsymbol{\Sigma}_e$ symmetric and positive definite, and $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbb{I})$ are the structural shocks, so that $\boldsymbol{\Sigma}_e = \boldsymbol{B}_0 \boldsymbol{B}_0'$. I set $p = 2$. Equations (1) and (2) form the quantile factor-augmented VAR model (QFAVAR) of Korobilis and Schröder (2025), which extends the factor-augmented VAR of Bernanke et al. (2005) by introducing quantile- and variable-specific latent factors. The vector $\boldsymbol{\mathcal{G}}_t$ plays a dual role: it conditions the country quantiles in the measurement equation (1) and evolves jointly with the quantile factors in the state equation (2).

The resulting model is high-dimensional. With 21 countries, 3 quantile factors, and 7 global factors, the measurement equation alone contains 504 loading parameters, while the state equation comprises 200 VAR coefficients, for a total of 704 parameters to be estimated from 312 monthly observations (January 2000–December 2025). Classical estimation of a model of this size is not feasible. To this end, Korobilis and Schröder (2025) estimate the model in a Bayesian framework that addresses the curse of dimensionality by placing horseshoe priors on the factor loadings and on the VAR coefficients, and I follow their approach. In this setting, the prior lets the data decide which parameters matter. In the measurement equation, it shrinks towards zero the loadings of countries whose inflation quantiles do not respond to a given global variable, while leaving the relevant loadings largely unrestricted. In the state equation, it

[4] Appendix A describes the sources and transformations of all series. Energy prices are converted from US dollars to euros before computing growth rates, and daily series are averaged within each month. EONIA is adjusted by the fixed spread of 8.5 basis points that links it to the €STR, so that the two segments of the overnight rate are comparable.

does the same for the many lags and cross-variable effects that carry little information, so the dynamics are driven by the few coefficients the data support.

The geopolitical disruption shock to Russian oil supply, $\varepsilon_t^{\mathcal{S}}$, is identified recursively, with $\boldsymbol{B_0}$ the lower-triangular Cholesky factor of $\boldsymbol{\Sigma}_e$ and $\mathcal{S}_t$ ordered firs in $\boldsymbol{y_t}$. The identifying assumption is a timing restriction: within the month, the geopolitical indicator does not respond to the other structural shocks in the system, while energy prices, financial and real conditions, monetary policy and the quantile inflation factors can all respond on impact to the geopolitical shock. The restriction is plausible because the events captured by $\mathcal{S}_t$, such as military escalations, sanctions and attacks on production and export infrastructure, reflect political and military decisions rather than contemporaneous economic developments (Caldara and Iacoviello, 2022). The same timing assumption is adopted by Caldara and Iacoviello (2022), Caravello and Martínez-Bruera (2024) and Brignone et al. (2026). Two implications are worth stressing. First, because $\mathcal{S}_t$ is ordered first, the identified shock and its impulse responses do not depend on how the remaining variables are ordered, and none of the other innovations needs a structural interpretation. Second, the shock is the component of $\mathcal{S}_t$ that cannot be predicted from the past of all variables in the system, so a gradual build-up of tensions already reflected in past energy prices or inflation is not counted as a shock. The shock is normalized to raise $\mathcal{S}_t$ on impact by one standard deviation of the index.

An attractive feature of the model is that the measurement equation (1) delivers country-level responses. Given the impulse responses of the state vector $\boldsymbol{y}_t$, the response of the q-th quantile of inflation in country i at horizon h follows from (1),

$$\phi_{i,h}^{q} = \frac{\partial \mathcal{Q}_q[\pi_{i,t+h}]}{\partial \varepsilon_t^{\mathcal{S}}} = \gamma_{i(q)} \frac{\partial \Pi_{t+h,(q)}}{\partial \varepsilon_t^{\mathcal{S}}} + \boldsymbol{\psi}_{i(q)} \frac{\partial \boldsymbol{\mathcal{G}}_{t+h}}{\partial \varepsilon_t^{\mathcal{S}}}, \qquad h = 0, 1, \ldots, H. \tag{3}$$

The response combines an indirect effect, through the common quantile factor, and a direct effect, through the global factors. It is computed draw by draw from the posterior of the loadings and of the VAR parameters, so the credible bands reflect uncertainty in both equations. Since inflation is standardized, I multiply $\phi_{i,h}^{q}$ by the sample standard deviation of π_{it} to express the responses in percentage points.

The responses of the individual quantiles describe how the shock moves each part of the inflation distribution, but not how it changes its shape. To capture the latter, I use two measures of inflation risk. Upside risk is the distance between the 90th percentile and the median, and downside risk is the distance between the median and the 10th percentile. A shock that shifts the whole distribution by the same amount moves

all quantiles together and leaves both measures unchanged. A shock that moves the 90th percentile by more than the median raises upside risk, so high-inflation outcomes become more likely relative to the center of the distribution. Similarly, a shock that lowers the 10th percentile by more than the median raises downside risk. The two measures add up to the distance between the 90th and the 10th percentiles, so together they capture changes in dispersion, while separately they indicate the side of the distribution where the change occurs. I compute their responses draw by draw, as differences between the impulse responses of the corresponding quantiles, so that the credible bands account for the joint uncertainty across quantile levels. For the Euro Area, the measures are built from the responses of the quantile factors $\mathbf{\Pi}_t$; for each member state, from the country-level responses $\phi^q_{i,h}$ in (3).

3 Main Results

This section presents the main results on how a geopolitical disruption to Russian oil supply reshapes the distribution of inflation in the Euro Area and across its member states. I first examine the responses of the Euro Area inflation quantiles and of the associated measures of risk, together with the responses of the global factors, which help interpret the nature of the shock. I then turn to the member states and study how the responses vary across countries and whether this variation is related to their exposure to Russian oil.

3.1 Does a Geopolitical Disruption Shock to Russian Oil Supply Raise or Lower Inflation?

Following Caldara et al. (2025), I first ask whether a geopolitical disruption to Russian oil supply raises or lowers inflation in the euro area. Such a shock can work through two opposing channels. Through the supply channel, it raises the expected scarcity and price of energy, pushing up firms' costs and headline inflation, an effect that precautionary demand for oil can amplify even before any physical disruption occurs. Through the demand channel, the same escalation raises uncertainty, tightens financial conditions and, for an energy importer such as the euro area, erodes real incomes, depressing consumption and investment and pulling inflation down. The two channels need not affect all parts of the inflation distribution in the same way. To the extent that the supply channel generates large cost increases in some months but

not in others, it should mostly affect the upper tail, whereas weaker demand should mostly depress the center and the lower tail.

Figure 2 reports the impulse responses of the 10th, 50th, and 90th percentile factors of euro area inflation. On impact, the shock moves the percentiles in different directions. The 90th percentile rises by about 0.2 percentage points, whereas the median falls slightly and the 10th percentile barely moves. Within a few months, however, all three percentiles increase, reaching a peak of about 0.25-0.3 percentage points after five to six months, and the effects fade after about 13 to 14 months. The shock therefore raises inflation across the distribution. The supply channel dominates, although the evidence below indicates that the demand channel is also at work.

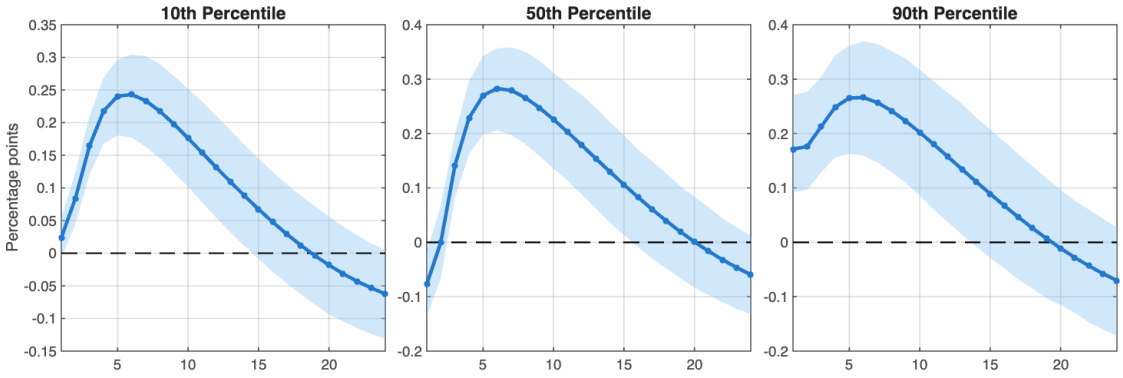


Figure 2: Dynamic Responses of Euro Area Inflation Quantiles to a Geopolitical Disruption to Russian Oil Supply

Note: Impulse responses of the tenth, fiftieth, and ninetieth percentile factors of euro area HICP inflation to a geopolitical disruption to Russian oil supply, estimated with the QFAVAR. The shock is identified recursively, with the Russian geopolitical oil supply disruption index of Iacoviello and Tong (2026) ordered first. Responses are scaled to a shock that raises the index by one standard deviation on impact and are expressed in percentage points, using the cross-country average standard deviation of inflation. Solid lines denote posterior means; shaded areas denote the 16th-84th percentile posterior credible bands.
Sample: January 2000 – December 2025.

The timing of the responses is informative. The shock moves the upper tail first. On its arrival, high inflation outcomes become more likely while the center of the distribution, if anything, moves down. The center and the lower tail catch up only after a few months, once higher energy costs pass through to a broader range of prices.

The responses of the global factors help disentangle the two channels, because the channels predict opposite signs for some variables and the same sign for others. Table

1 summarizes these predictions. A supply-driven disruption raises energy prices and supply chain pressures, whereas a contraction in demand lowers both. Both channels, instead, raise unemployment. Higher energy costs reduce output, and so does weaker spending. What distinguishes them is the comovement of activity and prices. Under the demand channel, unemployment rises while inflation falls; under the supply channel, unemployment and inflation rise together.

Table 1: Supply and Demand Channels: Expected and Estimated Signs

	Expected sign		
	Supply channel	Demand channel	Estimated response
Brent oil price inflation	$+$	$-$	0, then $-$
Gas price inflation	$+$	$-$	$+$
GSCPI	0 / $+$	$-$	$-$
CISS	0 / $+$	$+$	$+$
Unemployment rate	$+$	$+$	$+$ (delayed)
Shadow rate	$+$ / 0	$-$	$+$
Inflation	$+$	$-$	$+$

Note: The table reports the sign of the response of each variable expected under the supply channel and under the demand channel, and the sign of the estimated response to a geopolitical disruption to Russian oil supply (Figures H3 and 2). $+$: increase; $-$: decrease; 0: no significant response. The variables for which the expected signs differ discriminate between the two channels. Under the supply channel, the response of the shadow rate depends on whether monetary policy tightens against the rise in inflation ($+$) or looks through it (0), and financial stress and supply chain pressures may rise only indirectly ($0/+$), so these two variables discriminate only weakly. Unemployment rises under both channels and therefore does not discriminate between them; the channels are distinguished by the comovement of activity and prices. The GSCPI is constructed to purge demand factors from its components (Benigno et al., 2022), so its decline is consistent with, but not conclusive evidence of, weaker demand. Energy prices enter as year-on-year growth rates, so a temporary rise in the price level shows up as positive growth for about a year, followed by negative growth once the price returns toward its earlier level.

The last column of Table 1 reports the signs off the estimated responses, which are shown in full in Figure H3 in Appendix H.1. The evidence points to both channels being at work. The responses that discriminate most clearly match the supply channel. Gas price inflation rises by about 20 percentage points on impact and returns to zero after about five months, and inflation and unemployment both increase. Monetary policy tightens. The shadow rate rises significantly after a few months and peaks at about 0.15 percentage points after about a year, consistent with a central bank responding to inflationary pressures rather than to a fall in demand. Other responses

match the demand channel. Financial stress rises on impact and remains elevated for more than a year, and supply chain pressures ease. The two channels differ in timing. The rise in energy prices and inflation comes first, whereas unemployment increases only gradually and becomes significant after about a year, when inflation is already returning to its pre-shock level. Brent oil price inflation does not respond significantly on impact, suggesting that for crude oil the supply and demand effects largely offset, or that disrupted Russian oil is re-routed within an integrated global market. At longer horizons, both oil and gas price inflation turn negative. For gas, this largely reflects the year-on-year measurement: once the initial increase in prices is reversed, prices are compared with the elevated level of a year before. For oil, which does not rise on impact, the decline is instead consistent with weaker demand and tighter monetary policy in the second year after the shock.

3.2 Does the Shock Raise Upside or Downside Inflation Risk?

The percentile responses already suggest that the shock reshapes the inflation distribution rather than shifting it uniformly. The risk measures make this precise. An increase in upside risk, the distance between the 90th percentile and the median, means that high-inflation outcomes become more likely relative to the central one, so that the risk of an inflation surge rises. An increase in downside risk, the distance between the median and the 10th percentile, means the same for low-inflation outcomes. Figure 3 reports the responses of the two measures.

The two measures move in opposite directions on impact. Upside risk increases by about 0.25 percentage points, while downside risk declines by about 0.10 percentage points, as the 90th percentile rises and the median falls slightly. On impact, the shock therefore stretches the upper part of the distribution and compresses the lower part. The rise in upside risk is consistent with the supply channel operating through gas prices: it remains significant for about three months and dissipates after about five, in line with the transitory response of gas price inflation. The fall in the median, and hence in downside risk, is instead consistent with the demand channel, as financial stress rises on impact.

At longer horizons, upside risk returns to zero, while downside risk turns positive after about four months. This response is small, at most about 0.05 percentage points, and only marginally significant. Once the initial effect on the upper tail has faded, the three percentiles therefore move broadly together, and the shock shifts the

distribution rather than changing its shape. The effect of the shock on inflation risk is thus concentrated on impact and in the upper tail: it sharply raises the likelihood of high-inflation outcomes when the disruption occurs, and leaves the balance of risks broadly unchanged thereafter.

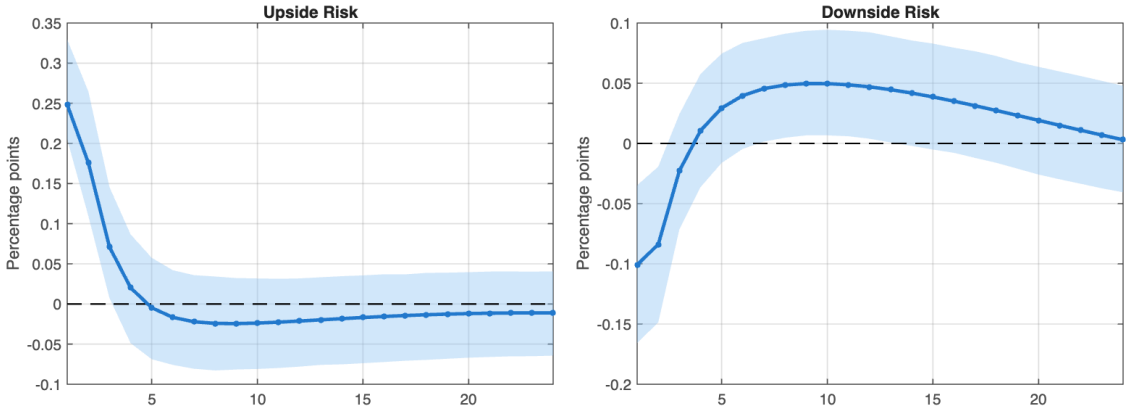


Figure 3: Dynamic Responses of Euro Area Inflation Tail Risks to a Geopolitical Disruption to Russian Oil Supply

Note: Impulse responses of upside risk and downside risk of euro area inflation to a geopolitical disruption to Russian oil supply, estimated with the QFAVAR. Risk measures are computed draw by draw from the responses of the quantile factors. The shock is identified recursively, with the Russian geopolitical oil supply disruption index of Iacoviello and Tong (2026) ordered first, and is scaled to raise the index by one standard deviation on impact. Responses are expressed in percentage points, using the cross-country average standard deviation of inflation. Solid lines denote posterior means; shaded areas denote the 16th–84th percentile posterior credible bands.
Sample: January 2000 – December 2025.

3.3 How Does the Response Differ across Member States?

The aggregate response masks considerable variation across member states. Figure 4 traces the shock through the 10th, 50th, and 90th percentiles of each member state's inflation, and Figure 5 reports the corresponding upside and downside risk measures. Three results are worth noting.

First, the shock is inflationary everywhere. Within a few months, all three percentiles rise in every member state. Negative responses on impact occur in some countries, most markedly in Bulgaria, Slovakia, and Slovenia, where all three percentiles fall, and in Latvia, as well as at the 10th percentile in Cyprus, Greece, Ireland,

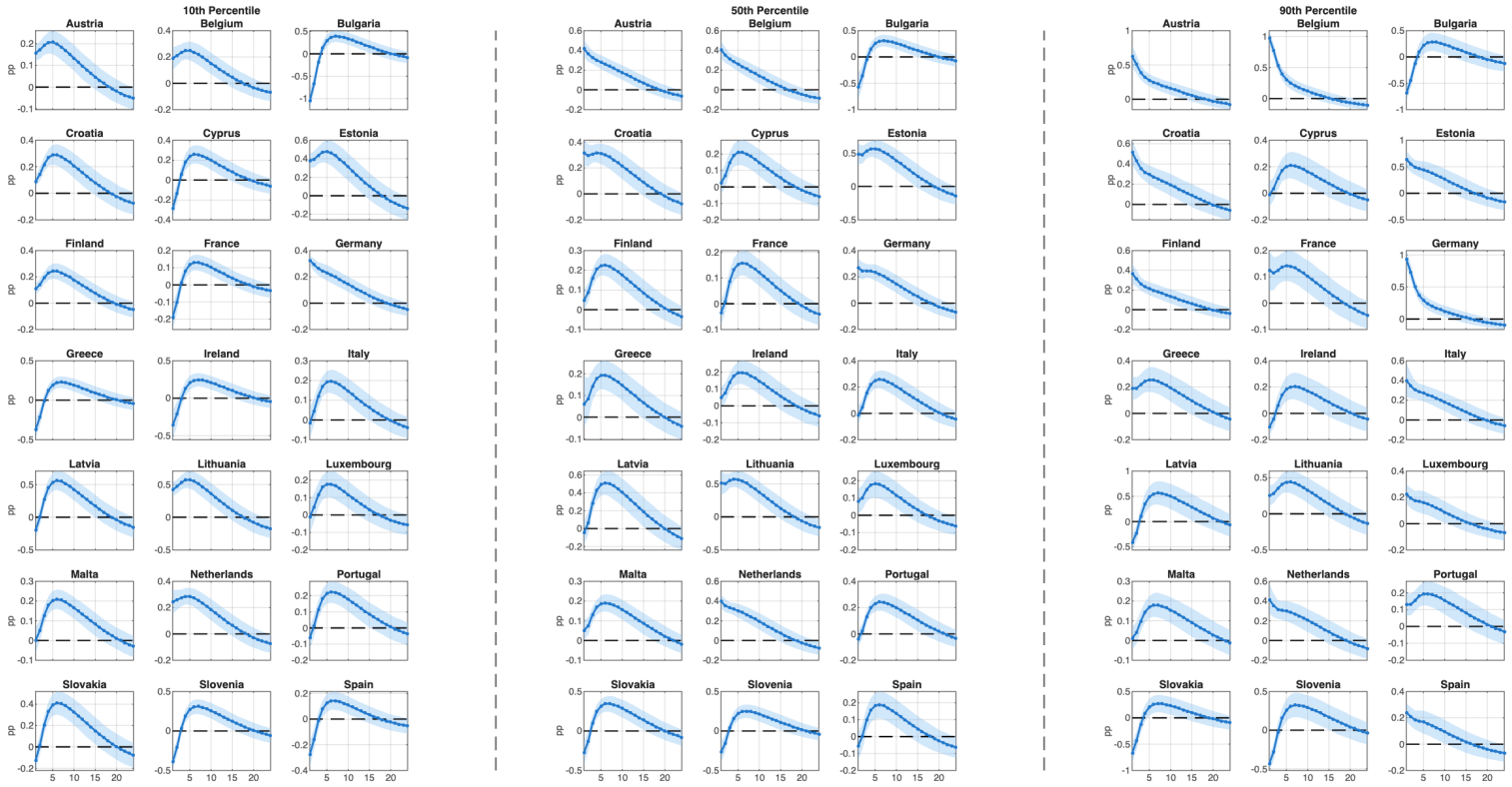


Figure 4: Dynamic Responses of Country-Level Inflation Quantiles to a Geopolitical Disruption to Russian Oil Supply

Note: Impulse responses of the tenth, fiftieth and ninetieth percentiles of the conditional distribution of country-level HICP inflation to a one standard deviation shock to the Russian geopolitical oil supply disruptions index. Solid lines denote posterior means; shaded areas denote 16th–84th percentile posterior credible bands. Sample: January 2000 – December 2025.

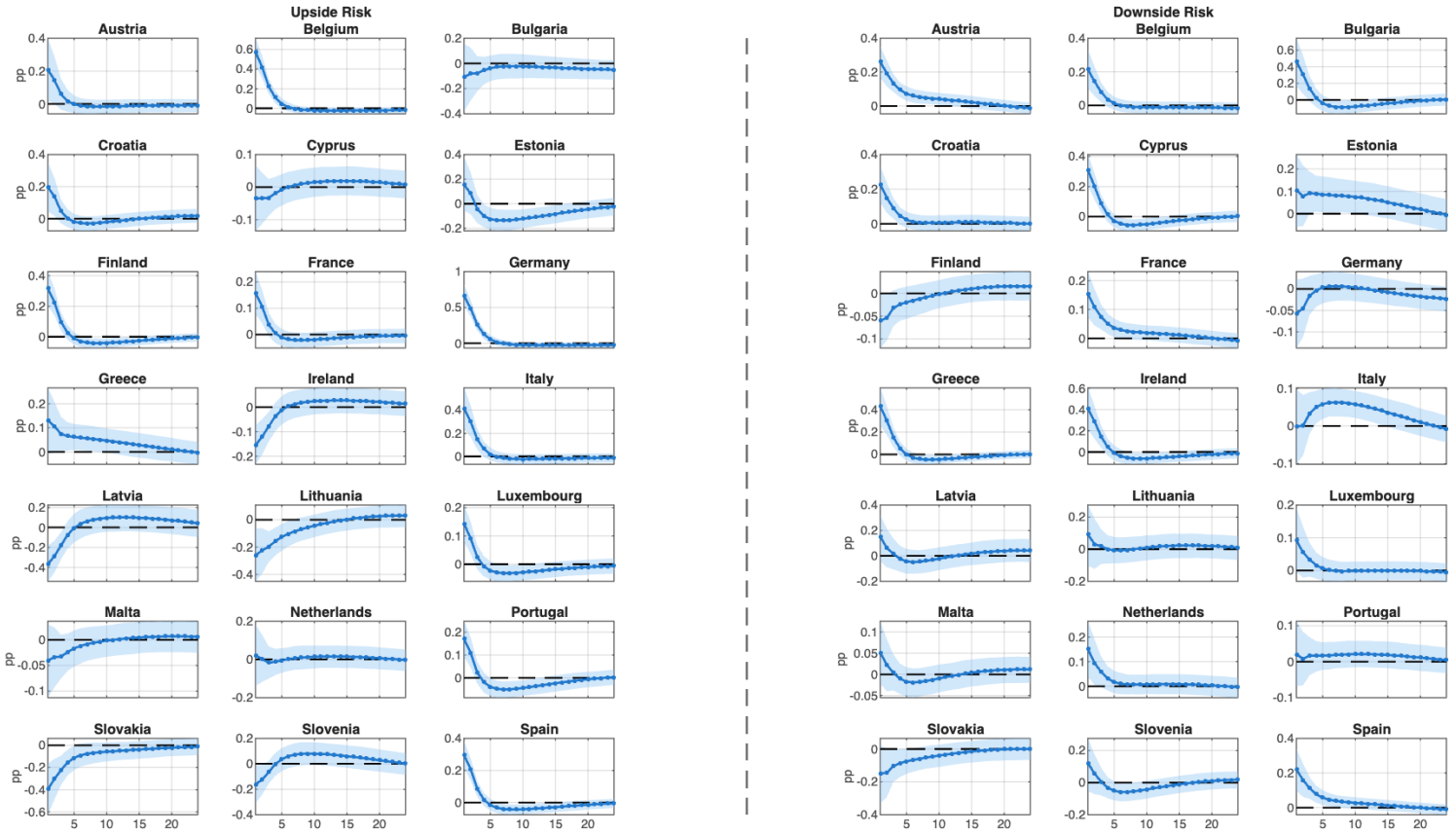


Figure 5: Dynamic Responses of Country-Level Upside and Downside Risks to a Geopolitical Disruption to Russian Oil Supply

Note: Impulse responses of upside inflation risk and downside inflation risk to a one standard deviation shock to the Russian geopolitical oil supply disruptions index, for each of the 21 member states. Solid lines denote posterior means; shaded areas denote 16th–84th percentile posterior credible bands. Note the differing vertical scales across panels. Sample: January 2000 – December 2025.

Spain, and France and at the 90th percentile in Ireland. They are short-lived and reverse within two to three months. The area-wide result therefore does not hinge on a few member states but reflects a broad-based disturbance.

Second, the timing and shape of the response differ across member states. In one group, which includes Germany, Belgium, Austria, the Netherlands, Italy, Croatia, Finland, Spain, and Luxembourg, the response is immediate and concentrated in the upper tail. The 90th percentile jumps on impact, by about 0.9 percentage points in Germany and Belgium and by 0.4–0.6 percentage points in Austria, Croatia, the Netherlands, and Italy, and declines steadily thereafter. In a second group, which includes Bulgaria, Slovakia, Slovenia, Latvia, Cyprus, Greece, Ireland, Malta, Portugal, and France, the response instead builds up gradually: impact effects are small or negative, and all three percentiles then rise together, peaking after about five to six months. Lithuania and Estonia combine the two patterns. The 10th percentile and the median rise already on impact and increase further for several months, while in Estonia the 90th percentile also jumps on impact. The largest peak responses of the 10th percentile and the median are found in the Baltic states, at about 0.5–0.6 percentage points. Because responses are expressed in percentage points of each country's own inflation, part of this dispersion reflects differences in inflation volatility (Table D3) as well as in the estimated loadings.

Third, the risk measures mirror these patterns. Upside risk increases significantly on impact in the first group, most strongly in Germany and Belgium (about 0.6 percentage points), followed by Italy, Finland, Spain, Austria, and Croatia, and returns to zero within about four to five months. In Slovakia, Latvia, Lithuania, Slovenia, and Ireland, by contrast, upside risk falls on impact, as the 90th percentile rises less than the median or initially declines. Downside risk increases on impact in most member states, most strongly in Bulgaria, Greece, Ireland, and Cyprus, where the 10th percentile falls before rising, but in most member states this effect also fades within a few months.[5]

The response therefore differs across member states. In Germany, Belgium, and the other member states of the first group, the shock produces an immediate but short-lived increase in upside risk. In the Baltic states, it produces a later and larger shift of

[5] At the country level, downside risk increases on impact in most member states, whereas it falls for the euro area as a whole. The euro area measures are built from the responses of the common quantile factors, while the country responses in (3) also include the direct effect of the global factors through the loadings $\psi_{i(q)}$.

the whole distribution, and in Bulgaria, Slovakia, and Slovenia a similar but smaller one, with upside risk falling or barely moving on impact.

3.4 Does Reliance on Russian Energy Explain the Differences?

The aggregate evidence shows that the shock operates through two channels. The supply channel, transmitted through higher energy prices, stretches the upper tail of the inflation distribution on impact and raises inflation across the distribution within a few months. The demand channel is visible in the prolonged rise in financial stress, the easing of supply chain pressures, and the gradual increase in unemployment. Nothing requires the two channels to have the same strength in every member state. Countries differ in how much energy they consume and where it comes from, which governs the strength of the supply channel, and in how incomes, labor markets, and financial conditions react to an energy shock, which governs the strength of the demand channel.

The country-level responses in Figure 4 are consistent with this view. In Germany, Belgium, and the other member states of the first group, the response resembles the aggregate one on impact: the upper tail jumps and upside risk rises briefly. In the Baltic states, the response builds up more slowly but extends to the whole distribution and reaches larger peaks, consistent with a gradual pass-through of energy costs to a broad range of prices. A common shock thus reshapes the inflation distribution differently across member states, which raises an obvious question: *what determines how strongly, and in which part of the distribution, each member state responds?*

The most natural candidate is exposure to the source of the disruption, which governs the strength of the supply channel. A geopolitical disruption to Russian oil supply should matter more for inflation in countries that relied more heavily on Russian energy and in which energy accounts for a larger share of consumption. The index captures disruptions to oil and, more broadly, to energy supply, and the shock raises European gas prices (Figure H3), so I consider reliance on both Russian oil and Russian natural gas. I capture these dimensions in a single measure of exposure,

$$\text{Exposure}_i = 100 \times \left(w_i^{oil}\, d_i^{oil} + w_i^{gas}\, d_i^{gas} \right), \tag{4}$$

where w_i^{oil} is the weight in country i's HICP of oil-based energy products (liquid fuels and fuels for personal transport), w_i^{gas} is the weight of gas, and d_i^{oil} and d_i^{gas} are the shares of Russia in the country's net imports of oil and petroleum products and

of natural gas. The import shares are measured by country of origin, so that imports transiting through another member state are attributed to the original supplier. The measure approximates the percentage of each country's consumer basket accounted for by oil and gas of Russian origin. It is zero for a country that imports neither from Russia, however energy-intensive its consumption, and it rises with both the importance of each fuel in the basket and the reliance on Russian supply of that fuel. To avoid capturing the reallocation of imports that followed the sanctions, I measure exposure before the invasion. All components refer to 2020, the last year for which gas import shares by country of origin are available, so that the weights and the import shares of both fuels are measured in the same year.[6]

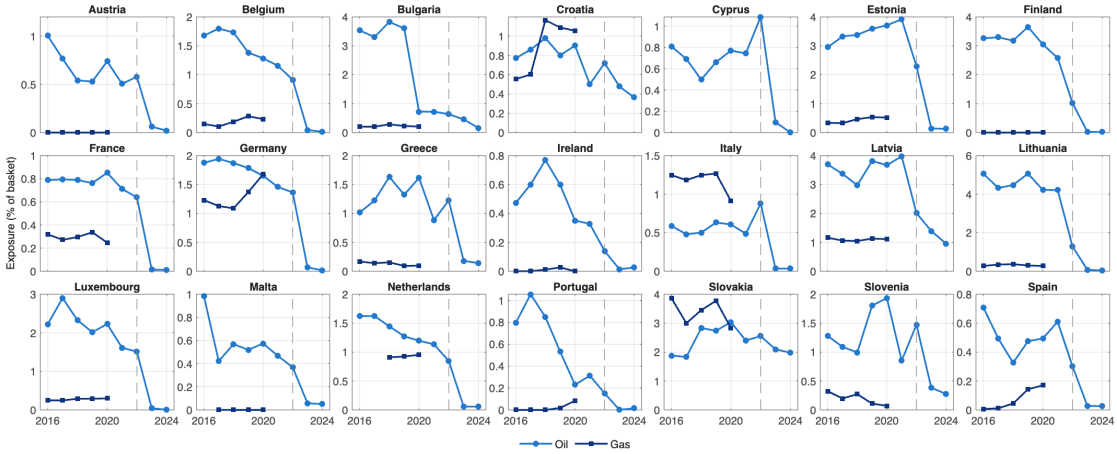


Figure 6: Exposure to Russian Oil and Gas by Member State, 2016–2024

Note: Exposure to Russian oil is the weight of liquid fuels and fuels for personal transport in the national HICP multiplied by the share of Russia in net imports of oil and petroleum products. Exposure to Russian gas is the weight of gas in the national HICP multiplied by the share of Russia in net imports of natural gas. Both are expressed in percent of the consumer basket, and import shares are measured by country of origin. Gas import shares are available until 2020. The dashed vertical line marks 2022.
Source: Eurostat; author's calculations.

Figure 6 shows that, before the war, exposure differed widely across member states.

[6]The measure does not include electricity, part of which is generated from natural gas, because the share of gas in power generation, and the extent to which it determines electricity prices, varies across countries. For Austria, the origin-based statistics report no imports of Russian gas, which likely reflects its role as a transit hub. Attributing most of Austria's gas imports to Russia would roughly double its exposure, but would leave it in the lower half of the distribution. The measure should therefore be read as a proxy for reliance on Russian energy rather than as a complete measure of it.

Combining the two fuels, it was highest in Slovakia (5.9% of the consumer basket), Latvia (4.8%), Lithuania (4.5%), and Estonia (4.2%), followed by Germany (3.3%) and Finland (3.1%), and lowest in Portugal, Ireland, Malta, Spain, and Austria, at less than 0.8%. The composition of exposure also differed. In the Baltic states and Finland, exposure came mainly from oil, whereas in Slovakia and Germany gas accounted for about half of it, and in Italy and Croatia for more than half. The two dimensions of the measure both matter: Finland relied heavily on Russian gas, but since gas has no weight in the Finnish consumer basket, its exposure came entirely from oil; Italy, by contrast, imported relatively little Russian oil, but its exposure to Russian gas was about one and a half times its exposure to Russian oil. After the war, exposure to Russian oil collapsed in almost every member state, in several cases after a temporary increase in 2022, before the embargoes took effect. The main exception is Slovakia, whose pipeline imports of crude oil were exempted from the embargo and whose exposure remained high.

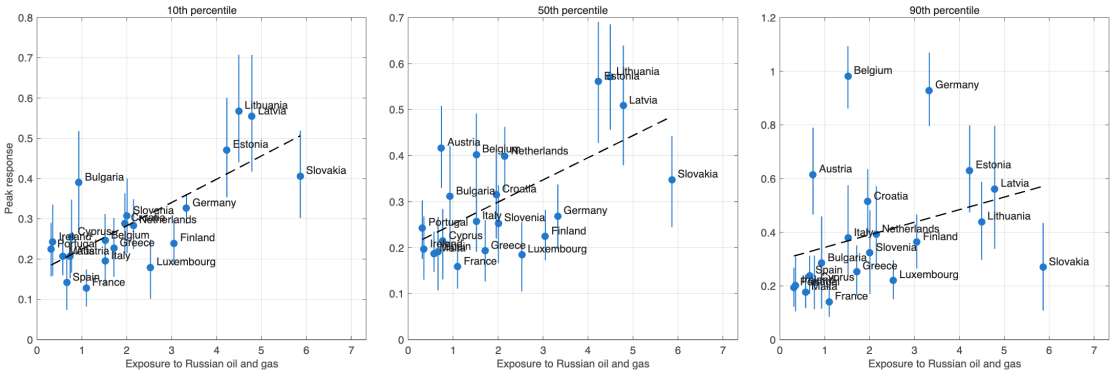


Figure 7: Exposure to Russian Oil and Gas and Peak Responses of Inflation Percentiles

Note: Each dot is a member state. The horizontal axis reports pre-war exposure to Russian oil and gas in 2020, in percent of the consumer basket, as defined in equation (4). The vertical axis reports the peak response of the 10th, 50th, and 90th percentile of the member state's inflation to a geopolitical disruption to Russian oil supply that raises the index by one standard deviation on impact, in percentage points, taken at the horizon at which the posterior median response is largest; vertical lines denote the 16th–84th percentile posterior credible bands at that horizon. Dashed lines are OLS fits through the posterior medians. Computed draw by draw, the slope has a posterior median of 0.058, 0.048, and 0.046 for the 10th, 50th, and 90th percentiles, and is positive in more than 99 percent of the draws. In OLS regressions on the posterior medians, the R^2 is 0.57, 0.37, and 0.10, and the p-values based on heteroskedasticity-robust standard errors are below 0.001, 0.009, and 0.14.
Sample: January 2000 – December 2025.

Figure 7 relates pre-war exposure to the peak response of each member state's inflation percentiles. More exposed member states tend to experience larger peak responses at all three percentiles, but the association is clearest for the 10th percentile and the median, where exposure accounts for a substantial part of the cross-country variation.

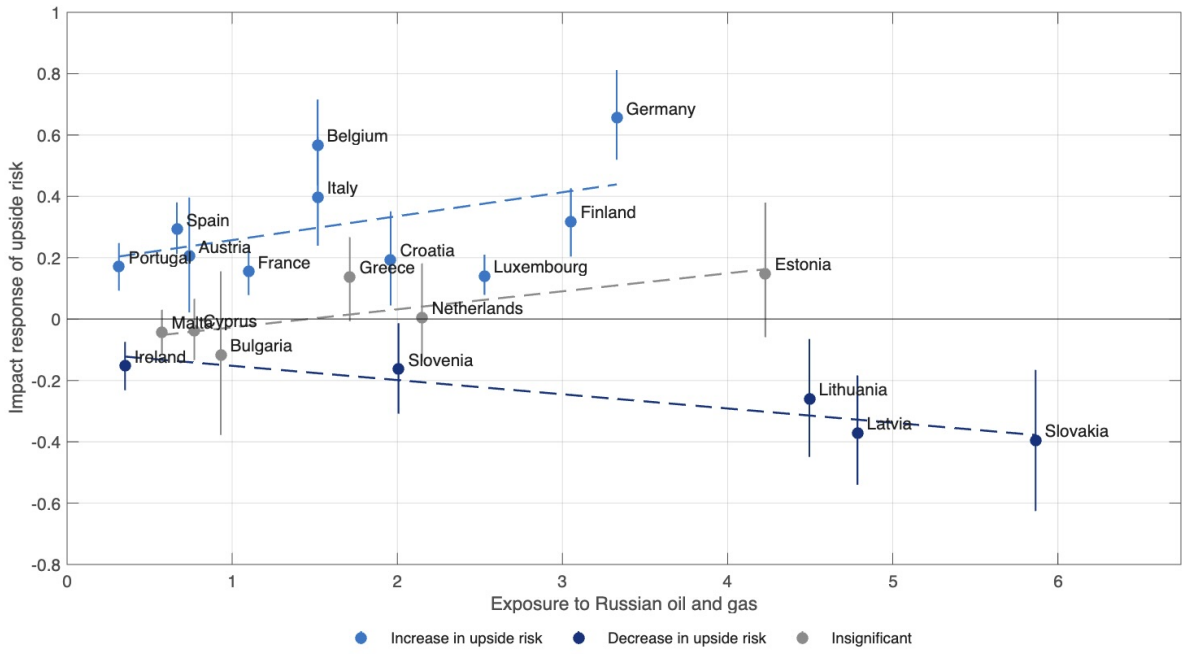


Figure 8: Exposure to Russian Oil and Gas and the Impact Response of Upside Risk

Note: Each dot is a member state. The horizontal axis reports pre-war exposure to Russian oil and gas in 2020, in percent of the consumer basket, as defined in equation (4). The vertical axis reports the impact response of upside risk (the difference between the 90th percentile and the median) to a geopolitical disruption to Russian oil supply that raises the index by one standard deviation on impact, in percentage points; dots denote posterior medians and vertical lines the 16th–84th percentile posterior credible bands. Member states are grouped according to whether the band lies above zero (increase in upside risk), below zero (decrease), or includes zero (insignificant). Dashed lines are OLS fits through the posterior medians within each group and are descriptive. Across all member states and computed draw by draw, the slope has a posterior median of -0.058, with a 68 percent credible band of $[-0.082, -0.032]$, and is negative in 99 percent of the draws.
Sample: January 2000 – December 2025.

The most exposed member states, the Baltic states, record the largest peak responses of the lower part of the distribution. In the upper tail, the association is weak. There,

the largest peaks occur in Belgium and Germany, two member states with moderate exposure whose 90th percentile jumps on impact, while Slovakia, the most exposed member state, records a comparatively small response of the 90th percentile, and Finland combines relatively high exposure with small responses at every percentile.

The impact response of upside risk shows the opposite pattern (Figure 8). Across member states, more exposed countries tend to experience a smaller increase in upside risk when the shock occurs, or a decrease. Upside risk rises on impact in ten member states, among them Germany, Belgium, Italy, and Finland, most of which have low or moderate exposure, whereas it falls in Slovakia, Latvia, Lithuania, Slovenia, and Ireland, including three of the four most exposed member states. The negative association across all member states therefore reflects mainly which of the two groups a member state belongs to. Within each group, exposure amplifies the response: among the member states in which upside risk rises, the increase is larger where exposure is higher, as in Germany and Finland, and among those in which it falls, the decline is larger where exposure is higher, as in Latvia and Slovakia. With only 21 member states, and with exposure correlated with other country characteristics, these patterns should be read as associations rather than causal effects. They suggest, nonetheless, that reliance on Russian energy is related to the size of the gradual, distribution-wide shift observed in the Baltic states, rather than to the immediate increase in upside risk in Germany, Belgium, and the other member states of the first group.

4 Additional Results

This section extends the main results in three directions. First, I re-estimate the model on samples that end before the invasion, to assess the role of the war period. Second, I construct a counterfactual path of inflation without the disruption shocks of 2022–2023, to measure their contribution to the inflation surge. Third, since the inflationary response emerges only once the war period is included, I allow the effects of the shock to depend on its size.

4.1 Was the 2022 Disruption Different?

The full-sample estimates combine disruptions of different magnitude. Before 2022, the largest realizations of the index are associated with the Russo-Georgian war of 2008 and the annexation of Crimea in 2014. As Figure C1 shows, these episodes are

registered as disruptions by the index, but at levels far below those reached in 2022, when the escalation of the war was accompanied by sanctions, embargoes, and a reorganization of the euro area's energy supply. To assess whether the results depend on this episode, I re-estimate the model on samples ending in December 2019 and December 2021, which exclude, respectively, both the pandemic and the invasion, and the invasion alone.[7]

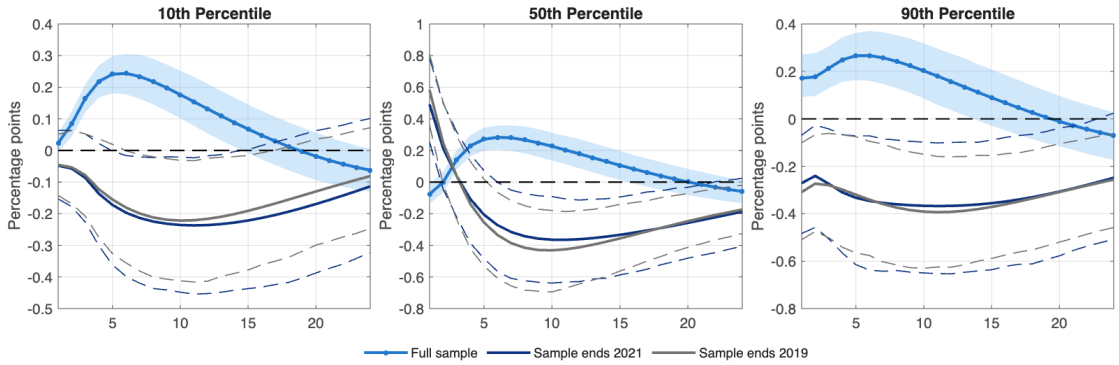


Figure 9: Responses of Euro Area Inflation Quantiles Across Estimation Samples

Note: Impulse responses of the 10th, 50th, and 90th percentile factors of euro area HICP inflation to a geopolitical disruption to Russian oil supply, estimated with the QFAVAR on the full sample (January 2000 – December 2025; light blue, shaded band) and on samples ending in December 2021 (dark blue) and December 2019 (gray), with dashed bands. The shock is identified recursively, with the Russia component of the geopolitical oil supply disruption subindex of Iacoviello and Tong (2026) ordered first, and raises the index by 60 points on impact in all samples, one standard deviation of the index in the full sample. Responses are expressed in percentage points, using the cross-country average standard deviation of inflation in each sample. Solid lines denote posterior means; bands denote the 16th–84th percentile posterior credible bands.

Figure 9 compares the responses of the euro area inflation percentiles in the full sample with those obtained in the two samples that end before the invasion. To make the three sets of estimates comparable, the shock is set to the same size in all samples, 60 index points, one standard deviation of the index in the full sample and the size used in the main results. The two pre-war samples deliver almost identical responses, which differ markedly from those of the full sample. Before the war, a geopolitical disruption to Russian oil supply is disinflationary. The 90th percentile falls from im-

[7] The shorter samples contain 240 and 264 monthly observations, against 312 in the full sample, while the number of parameters to be estimated is unchanged at 704. The subsample estimates therefore rely more heavily on the shrinkage imposed by the horseshoe priors and are less precise, in particular as regards the separation between the median and the tail quantile factors.

pact, and the median and the 10th percentile follow within a few months. All three percentiles reach a trough after about ten months, at about -0.2 percentage points for the 10th percentile and about -0.4 percentage points for the median and the 90th percentile, and remain below their pre-shock level for most of the following two years. On impact, the median instead rises while the 90th percentile falls, so that upside risk declines (Figure H4 in the Appendix), the opposite of the full-sample response. Because the shorter samples separate the median from the tail quantile factors less precisely, I interpret this impact split with caution.

The pre-war responses are consistent with the demand channel dominating. The disruptions captured by the index before 2022 were small compared with those of 2022 and did not lead to a lasting interruption of energy supply. Their effect on inflation therefore operated mainly through higher uncertainty and weaker demand rather than through higher energy costs. The responses of the global factors support this interpretation (Figure H5 in the Appendix). In the pre-war samples, financial stress rises by more than in the full sample, supply chain pressures and oil price inflation fall, and the shadow rate declines persistently, consistent with monetary policy easing in response to weaker demand, whereas in the full sample it rises. The inflationary effect documented in Section 3 thus emerges only once the war period is included, which raises the question of whether the effects of the shock depend on its size, the subject of Section 4.3.

The responses of upside risk on impact across member states, in Figure 10, tell the same story. In the full sample, the shock widens the upper tail on impact in about half of the member states, most strongly in Germany and Belgium. In the pre-war samples, this pattern disappears: upside risk falls on impact in most member states, and it rises significantly only in Portugal and Slovenia.[8] The increase in upside risk documented in the main results is therefore a feature of the 2022 disruption.

The contrast between samples points to the nature of the disruption as a key determinant of the response. Disruptions that remain limited, without a lasting interruption of energy supply, operate mainly through the demand channel, gradually depressing activity and prices and shifting the whole distribution down without raising upside risk. A disruption that interrupts supply operates through the supply channel, raising inflation across the distribution and, on impact, its upper tail. In the terms of

[8]The country-level estimates in the pre-war samples are imprecise, since the shorter samples identify the joint response of the median and the 90th percentile more accurately than their difference, and should not be interpreted individually.

Caldara et al. (2025), whether a geopolitical shock raises or lowers inflation depends on which of the two channels prevails, and the evidence here suggests that this in turn depends on whether the disruption affects actual energy supply. The full-sample estimates should accordingly be read as reflecting predominantly the 2022 episode. Because the model is linear, however, it cannot separate the role of the nature of the shock from that of its size, since the pre-war episodes were both smaller and of a different kind.

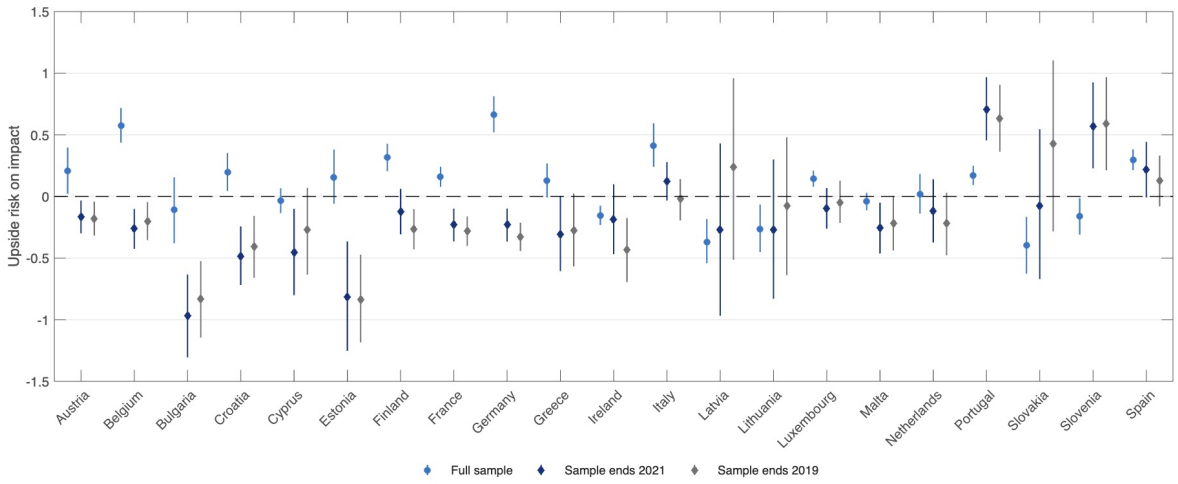


Figure 10: Upside Inflation Risk by Member State Across Estimation Samples

Note: Impact response of upside risk (90th percentile minus median of inflation) in each member state, in percentage points, estimated on the full sample (light blue circles) and on samples ending in December 2021 (dark blue diamonds) and December 2019 (gray diamonds). In all samples, the shock raises the index by 60 points on impact. Markers denote posterior means; vertical lines denote the 16th–84th percentile posterior credible bands.

4.2 What If There Had Been No Disruptions After the Invasion?

The previous subsection shows that the effects of the shock are identified mainly by the 2022 episode. A natural question is therefore how much of the inflation surge that followed the invasion can be attributed to the disruption. To answer it, I construct a counterfactual history in which the geopolitical disruption shocks are set to zero from February 2022 to December 2023, while all other shocks are kept at their estimated values. The contribution of the disruption is the difference between the fitted

and the counterfactual path of each country's inflation quantiles, computed for every posterior draw; the euro area figures are cross-country averages with equal weights.[9]

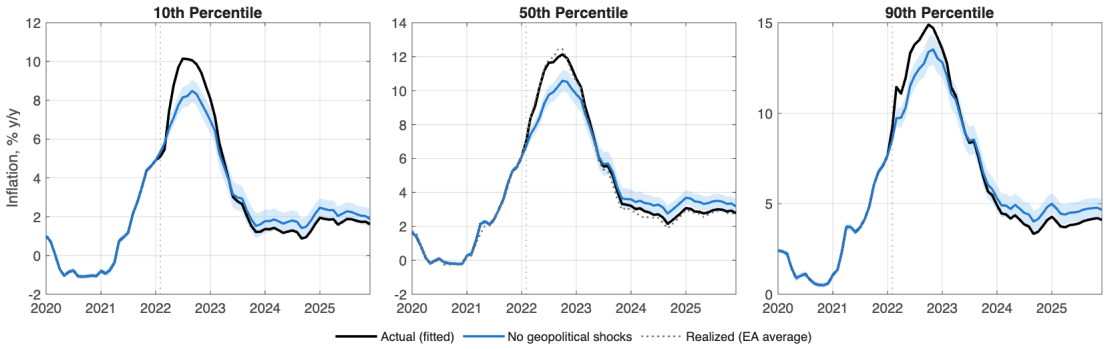


Figure 11: Euro Area Inflation Quantiles With and Without the Geopolitical Disruption

Note: Fitted 10th, 50th, and 90th percentiles of euro area inflation (black) and their counterfactual paths when the geopolitical disruption shocks are set to zero from February 2022 to December 2023 (light blue), with all other shocks held at their estimated values. Euro area values are cross-country averages of the member-state quantiles, with equal weights. The dotted line in the middle panel is realized inflation, averaged across member states. The vertical line marks February 2022. Solid lines denote posterior means; the shaded area denotes the 16th–84th percentile posterior credible band of the counterfactual.
Sample: January 2000 – December 2025.

Figure 11 compares the fitted and counterfactual paths of the euro area quantiles. The model tracks the surge closely: the fitted median follows realized inflation through the peak. Without the disruption, inflation would still have risen to rates not seen since the creation of the monetary union, but by less. The disruption raised all three quantiles, by about two percentage points at its peak in June and July 2022, a few months before inflation itself peaked. Averaged over 2022 and 2023, it raised the 10th percentile, the median, and the 90th percentile by 0.64, 0.73, and 0.73 percentage points. Over the window as a whole, the disruption therefore shifted the euro area distribution rather than changing its shape. In 2024 and 2025, the counterfactual lies above the fitted path of all three quantiles, by about half a percentage point, reflecting the reversal of energy prices and the tighter monetary policy that followed the disruption.

[9]The counterfactual removes all identified shocks in the window, positive and negative. Removing only the positive realizations would mechanically attribute a positive contribution to the disruption, since the shocks have mean zero. Details on the construction are reported in Appendix F.

The timing of the contributions differs across member states, consistent with the impulse responses (Figure H7 in the Appendix). In Germany, Belgium, and Austria, the effect on the upper tail came first: the contribution to the 90th percentile peaked in March 2022, immediately after the invasion, at about 5.7 to 8.8 percentage points, well above the contribution to the median. In the member states in which the response builds up gradually, the contribution was instead negative at the onset of the war, most markedly at the 10th percentile, and turned positive within two to three months, peaking in the summer of 2022. Because the country contributions peak in different months, and partly offset each other at the onset, the euro area contribution peaks later and at a lower level than in many member states.

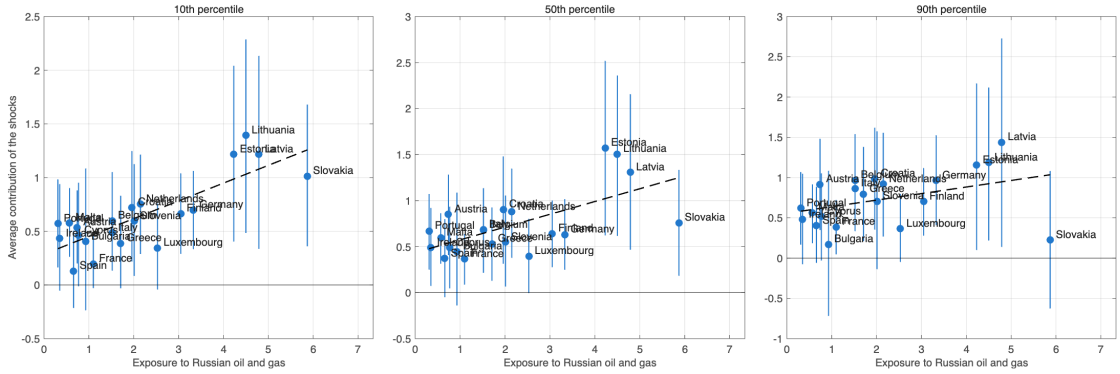


Figure 12: Contribution of the Geopolitical Disruption and Exposure to Russian Oil and Gas

Note: Each dot is a member state. The horizontal axis reports pre-war exposure to Russian oil and gas in 2020, in percent of the consumer basket, as defined in equation (4). The vertical axis reports the contribution of the geopolitical disruption shocks to the 10th, 50th, and 90th percentile of the member state's inflation, averaged from February 2022 to December 2023, in percentage points. Dots denote posterior means; vertical lines denote the 16th–84th percentile posterior credible bands. Dashed lines are OLS fits through the posterior means.
Sample: January 2000 – December 2025.

The contribution of the disruption varied considerably across member states. It was largest in the Baltic states, where it raised the median by about 1.3 to 1.6 percentage points on average over 2022 and 2023, roughly twice the euro area average, and smallest in Spain, France, and Luxembourg, where it remained around 0.4 percentage points. Figure 12 shows that these differences line up with pre-war exposure. At all three percentiles, member states more exposed to Russian oil and gas received a larger contribution, and the association is clearest at the 10th percentile and the median. The

main exception is Slovakia, the most exposed member state, whose contribution was only moderate at the median and small and imprecisely estimated at the 90th percentile. With only 21 member states, these patterns should again be read as associations. The counterfactual nonetheless confirms, from the observed history rather than from impulse responses alone, that the burden of the disruption fell disproportionately on the member states that relied most on Russian energy.

These figures should be read with two caveats. First, the counterfactual is computed with the linear QFAVAR, which scales the effects of the 2022 disruption from the average response to all disruptions in the sample. Since the nonlinear estimates of Section 4.3 indicate that large disruptions have more than proportional effects, the contribution of the 2022 disruption is, if anything, understated. Second, the counterfactual holds the coefficients of the model fixed, abstracting from changes in policy or behavior that the absence of the disruption might have induced.

4.3 Does the Size of the Disruption Matter?

The previous results raise a further question: why does the same index produce an inflationary response in the full sample but not before 2022? One possibility is that the effects of a disruption are not proportional to its size. A modest disruption may raise uncertainty and depress demand without threatening supply, whereas only a large disruption may move energy markets enough to trigger the supply channel. A linear model cannot capture this, because it scales the response to a large shock up from the average response to small ones.

To allow the response to depend on the size of the shock, I follow Brignone et al. (2026) and estimate a nonlinear VARX in two steps. First, I recover the structural disruption shock $\varepsilon_t^{\mathcal{S}}$ from the QFAVAR for each posterior draw. Second, for each draw, I estimate by OLS a VAR in the same variables as the state equation (2), with the index replaced by the identified shock and its square,

$$\boldsymbol{z}_t = \boldsymbol{c} + \sum_{j=1}^{2} \boldsymbol{D}_j \boldsymbol{z}_{t-j} + \boldsymbol{b}\, \varepsilon_t^{\mathcal{S}} + \boldsymbol{d}\, (\varepsilon_t^{\mathcal{S}})^2 + \boldsymbol{u}_t,$$

where $\boldsymbol{z}_t$ collects the global factors other than $\mathcal{S}_t$ and the three quantile factors, $\boldsymbol{z}_t = [\pi_t^{oil}, \pi_t^{gas}, \mathcal{F}_t, gscpi_t, u_t, R_t, \boldsymbol{\Pi}_t']'$. The impact response to a shock of size δ is $\boldsymbol{b}\delta + \boldsymbol{d}\delta^2$, and it is propagated through the autoregressive matrices.[10] In a linear model ($\boldsymbol{d} = \boldsymbol{0}$),

[10] Appendix G describes the computation of the impulse responses and of their credible bands.

the response per unit of shock is the same for every δ. A nonzero $\boldsymbol{d}$ instead implies that large disruptions have effects that are more, or less, than proportional to those of small ones.

Figure 13 compares the responses of the inflation percentiles, per 100 index points, to a typical shock of one standard deviation of the identified shock, about 40 index points, and to a large shock of nine standard deviations, about 360 points, close to the disruption identified in March 2022. In a linear model the two sets of responses would coincide. Instead, the large shock has effects that are far more than proportional to its size. Per 100 index points, it raises the peak response of the 10th percentile about four times as much as a typical shock, that of the median about three times as much, and that of the 90th percentile more than twice as much, with peaks reached after six to eight months. After a typical shock, the response of the 10th percentile is not significant, and that of the median is small and short-lived.

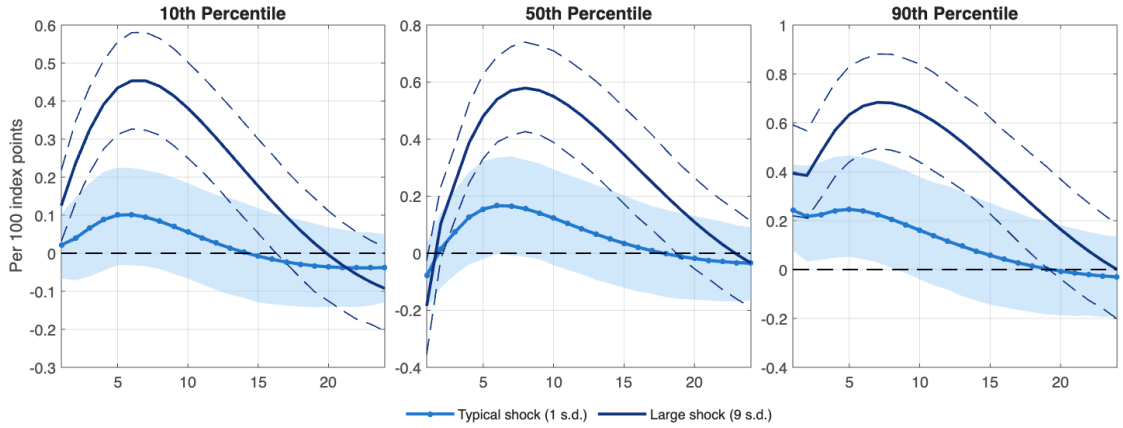


Figure 13: Responses to Small and Large Geopolitical Disruptions

Note: Responses of the 10th, 50th, and 90th percentile factors of euro area inflation, in percentage points, to a geopolitical disruption to Russian oil supply of 40 index points (one standard deviation of the identified shock; light blue) and 360 index points (nine standard deviations; dark blue), expressed per 100 index points, estimated with the nonlinear VARX described in the text. In a linear model the two responses would coincide. Lines denote posterior means; bands denote the 16th–84th percentile credible bands across QFAVAR posterior draws.
Sample: January 2000 – December 2025.

The source of the nonlinearity is the response of energy prices. The nonlinear term is clearly positive for gas price inflation and, less precisely, for oil price inflation, whereas for the inflation percentiles themselves the impact coefficients are imprecisely estimated: the size effect on inflation builds up over the following months, as the

larger response of energy prices propagates. Large shocks also differ in kind. The impact response of oil price inflation turns from negative to positive for shocks above about 140 index points, and that of supply chain pressures for shocks above about 320 points. These thresholds lie between the largest pre-war disruptions, of 85 to 100 points in March 2014 and August 2008, and the disruption of March 2022, of about 370 points. Below them, a disruption leaves oil prices and supply chains broadly unaffected or eases them, consistent with the demand channel; above them, it raises both, and the supply channel takes over.[11]

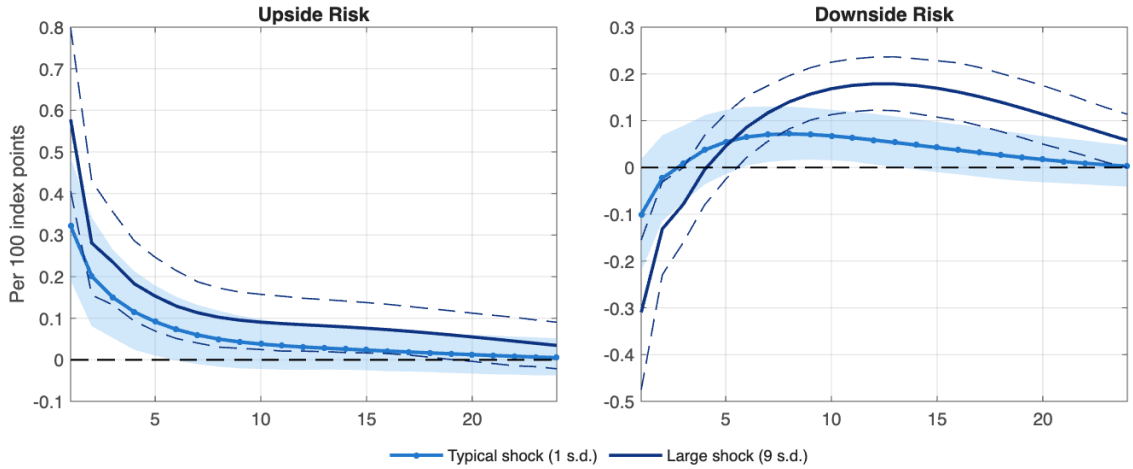


Figure 14: Inflation Risk After Small and Large Geopolitical Disruptions

Note: Responses of upside risk (90th percentile minus median) and downside risk (median minus 10th percentile) of euro area inflation to a geopolitical disruption to Russian oil supply of 40 index points (one standard deviation of the identified shock; light blue) and 360 index points (nine standard deviations; dark blue), expressed per 100 index points, in percentage points. Lines denote posterior means; bands denote the 16th–84th percentile credible bands across QFAVAR posterior draws. *Sample: January 2000 – December 2025.*

The consequences for inflation risk are shown in Figure 14. On impact, a large disruption raises upside risk by about 0.6 percentage points per 100 index points, against about 0.4 for a typical shock, and the increase fades more slowly. At the same time, it compresses the lower part of the distribution, lowering downside risk on impact. After a few months, the pattern reverses: downside risk rises, by about twice as much per index point after a large disruption as after a typical one, and remains elevated for

[11] The thresholds are not precisely estimated: the sign change occurs in about 70 and 80 percent of the posterior draws, respectively. Appendix H.5 reports the responses to shocks of the size of the main historical episodes.

more than a year. Large disruptions thus first stretch the upper tail and then widen the distance between the median and the lower tail, increasing the dispersion of inflation outcomes, consistent with the rise in inflation uncertainty associated with geopolitical risk by Caldara et al. (2025).

These results reconcile the subsample and full-sample evidence. Disruptions of the size observed before 2022 have modest effects on inflation and leave energy markets and supply chains broadly unaffected, consistent with the demand channel and with the absence of an inflationary response in the pre-war samples. A disruption of the size of 2022 activates the supply channel, raises energy prices sharply, and has effects on inflation and on its dispersion that are far more than proportional to its size. The nonlinearity is, however, identified mainly by the large shocks around the onset of the war, between February and April 2022, the only months in which the shock exceeds three standard deviations in absolute value, so the estimates should be read as describing how the 2022 disruption differed from earlier ones rather than as precise estimates of the thresholds.

5 Concluding Remarks

This paper has studied how geopolitical disruptions to Russian oil supply reshape the distribution of inflation across and within the euro area. Combining the AI-based index of Iacoviello and Tong (2026) with the quantile factor-augmented VAR of Korobilis and Schröder (2025), I trace an identified disruption shock through the 10th, 50th, and 90th percentiles of inflation in the euro area and in each of its 21 member states.

Four findings stand out. First, a disruption raises inflation across the distribution, but not uniformly. The upper tail moves first: on impact, upside risk rises sharply, and the median and the lower tail catch up only after a few months. Both a supply and a demand channel are at work: gas prices rise and monetary policy tightens, while financial stress rises, supply chain pressures ease, and unemployment increases gradually. Second, member states do not respond alike. In Germany, Belgium, and several other member states, the response is immediate and concentrated in the upper tail; in the Baltic states, it builds up more gradually but extends to the whole distribution and reaches larger peaks. These differences are associated with pre-war reliance on Russian oil and gas, especially in the lower part of the distribution. Third, the disruptions of 2022 and 2023 contributed substantially to the inflation surge, adding about two percentage points to euro area inflation at their peak, in mid-2022, and about three

quarters of a percentage point on average over the two years, with the largest contributions in the most exposed member states. Fourth, the inflationary effect is a feature of large disruptions. In samples that end before the invasion, disruptions are disinflationary. A nonlinear specification shows that only disruptions well above the size of the pre-war episodes raise oil prices and supply chain pressures, and that the effects of a disruption of the size observed in 2022 on inflation and on its dispersion are far more than proportional to its size.

These results carry three implications for policy in the euro area. First, the consequences of a geopolitical disruption are better described by its effect on the tails of the inflation distribution than by its effect on the mean: the immediate consequence of a large disruption is a higher probability of high inflation outcomes, which may matter for the anchoring of inflation expectations. Second, a single monetary policy faces heterogeneous inflation risks across member states, and the member states most reliant on the source of the disruption bear a disproportionate share of its effects. Monetary policy cannot address these differences, whereas reducing dependency on a single energy supplier, as pursued since 2022, is likely to reduce them. Third, the response to a geopolitical disruption is not proportional to its size. Disruptions that remain limited call for a different assessment from those that curtail energy supply, and monitoring the supply dimension of geopolitical events is therefore as important as monitoring their frequency.

The analysis has limitations that suggest directions for future research. The large effects are identified mainly by the 2022 episode, so the estimates describe how that disruption differed from earlier ones more than they pin down precise thresholds. The counterfactual relies on the linear model and, given the nonlinearity, likely understates the contribution of the 2022 disruption. The exposure measure is a proxy: it refers to 2020, the last year for which gas import shares by country of origin are available, it attributes no Russian gas to transit hubs such as Austria, and it does not capture indirect reliance through electricity. Separating disruptions that interrupt supply from those that only raise the threat of interruption, for instance using the distinction between geopolitical acts and threats, and extending the analysis to other commodities would help clarify when geopolitical events become inflationary.

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Online Appendix

Cutting the Flow: Geopolitical Disruptions to Russian Oil Supply and the Tails of Euro Area Inflation

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October 2026

Note to Readers: This online appendix is intended to complement the main paper and is not self-contained. Readers are encouraged to consult the main text for full context, methodological framework, and empirical results.

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A Data

All series are monthly and cover January 2000 to December 2025, for a total of 312 observations. Table A1 lists the variables, their sources, and their transformations. Series published at a daily frequency, namely the Brent oil price, the CISS, and the euro short-term rate, are converted to monthly frequency by taking averages over the month.

Global factors. The Russian geopolitical oil supply disruption index of Iacoviello and Tong (2026) enters in levels. Energy prices are converted from US dollars to euros by dividing the dollar price by the monthly average of the ECB reference exchange rate, expressed in US dollars per euro, so that they reflect the prices paid by euro area households and firms. They then enter as year-on-year growth rates, computed as $100 \times (P_t / P_{t-12} - 1)$, which requires price data from January 1999. The CISS, the GSCPI, and the unemployment rate enter in levels; the unemployment rate is seasonally adjusted at the source.

Monetary policy stance. The shadow short rate of Wu and Xia (2020) is available from September 2004 to December 2022. Outside this period, when policy rates were not constrained by the lower bound, the shadow rate essentially coincides with the overnight market rate, and I use the latter to complete the series. Before September 2004, I use the monthly average of EONIA. After December 2022, I use the monthly average of the euro short-term rate (€STR). Since October 2019, EONIA has been defined as €STR plus a fixed spread of 8.5 basis points; I subtract this spread from EONIA so that the two segments are measured on the same basis.

Inflation. Inflation is the year-on-year growth rate of the headline HICP for each of the 21 member states, as published by Eurostat. Year-on-year rates do not require seasonal adjustment.

Standardization. Before estimation, all series are standardized by subtracting their sample mean and dividing by their sample standard deviation, country by country for the inflation panel. Impulse responses are converted back to the original units of each variable by multiplying by the corresponding standard deviation, as described in Appendix E.

Exposure. The exposure measure of Section 3 combines annual HICP item weights with Russia's share in net imports of oil and of natural gas. The weights refer to gas (CP0452), and to liquid fuels (CP0453) and fuels for personal transport equipment (CP0722), which together form the oil-based component; they are published in per mille and divided by 1000. Import shares are measured by country of origin, which attributes imports that transit through another member state to the original supplier. Gas dependency is available until 2020, so the pre-war measure combines the 2021 weights and oil dependency with the 2020 gas dependency.

Table A1: Data sources

Variable	Description and transformation	Source	Link
Global factors			
Russian oil GPR	Russian geopolitical oil supply disruption index, level	Iacoviello and Tong (2026)	Link
Brent oil price inflation	Brent crude oil price, monthly average, converted to euros, year-on-year growth rate, percent	FRED	Link
European gas price inflation	Global price of natural gas, EU, converted to euros, year-on-year growth rate, percent	FRED	Link
Exchange rate	ECB euro foreign exchange reference rate, US dollar per euro, monthly average; used to convert oil and gas prices to euros	ECB	Link
CISS	Composite Indicator of Systemic Stress (new CISS), euro area, monthly average, level	ECB	Link
GSCPI	Global Supply Chain Pressure Index, standard deviations from average, level	FRB of New York	Link
Unemployment rate	Euro area unemployment rate, seasonally adjusted, percent	ECB	Link
Shadow short rate	Euro area shadow short rate, percent, September 2004 – December 2022; completed with the overnight rate outside this period	Wu and Xia (2020)	Link
EONIA	Euro overnight index average, monthly average, percent, minus 8.5 basis points; used before September 2004	ECB	Link

Table A1 continued

Variable	Description and transformation	Source	Link
€STR	Euro short-term rate, monthly average of daily rates, percent; used after December 2022	ECB	Link
Inflation			
HICP inflation	Headline inflation, year-on-year growth rate of the overall HICP, percent, 21 member states	Eurostat	Link
Exposure			
HICP item weights	Weights of gas (CP0452), liquid fuels (CP0453) and fuels for personal transport equipment (CP0722) in the national HICP, per mille, annual	Eurostat	Link
Russian oil dependency	Share of Russia in net imports of oil and petroleum products, by country of origin, percent, annual	Eurostat	Link
Russian gas dependency	Share of Russia in net imports of natural gas, by country of origin, percent, annual (available until 2020)	Eurostat	Link

B Imports of Russian Oil and Petroleum Products

Table B2: Imports of Russian Oil and Petroleum Products

	European Union (27 countries)		Euro Area (21 countries)	
	Thousand tonnes	Percentage change	Thousand tonnes	Percentage change
2020	169,929.911		131,617.699	
2021	173,229.010	1.94%	137,204.237	4.24%
2022	132,022.392	−23.79%	102,454.358	−25.33%
2023	21,403.751	−83.79%	10,089.959	−90.15%
2024	15,308.301	−28.48%	6,522.124	−35.36%

C Russian Geopolitical Oil Supply Disruptions Index

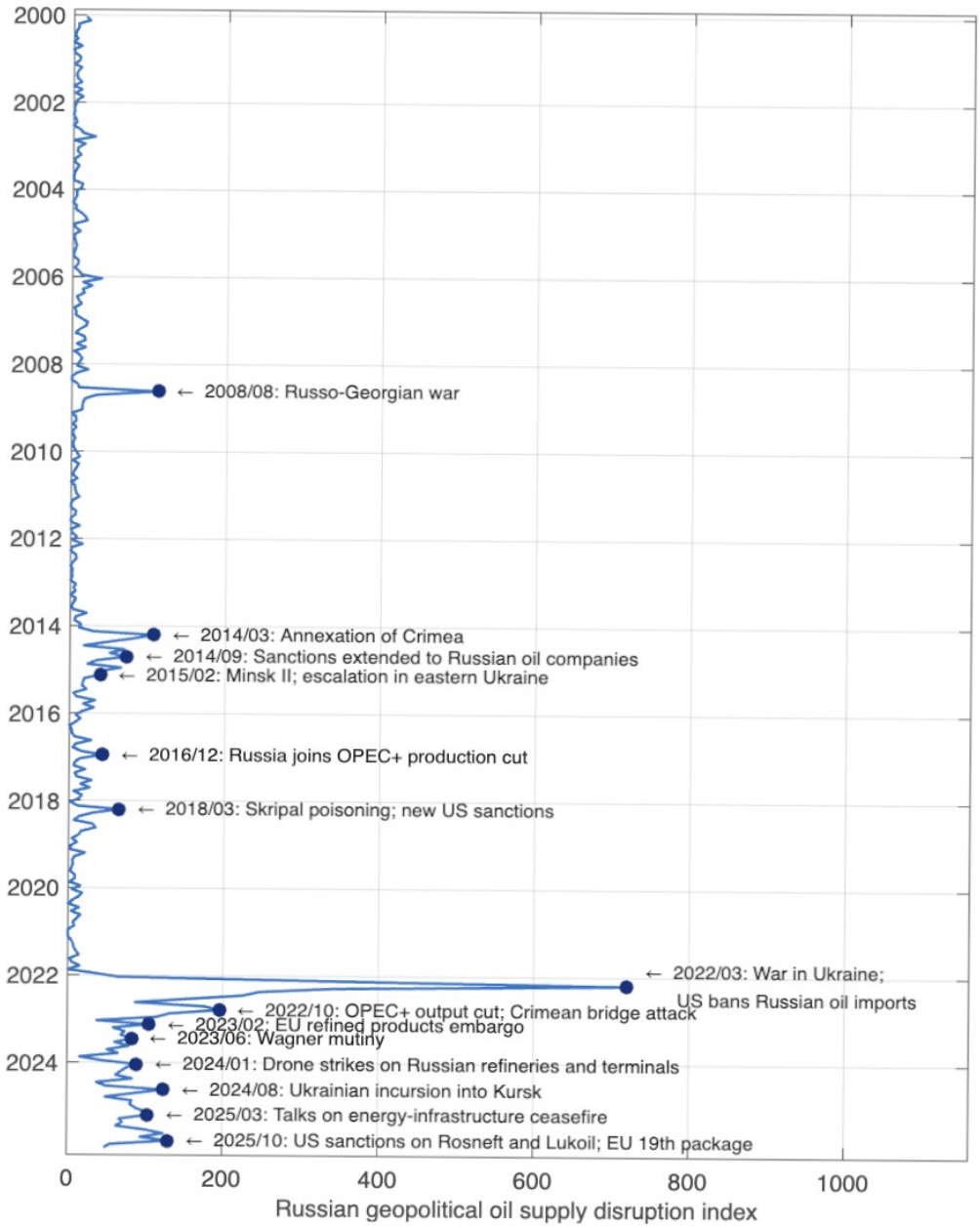


Figure C1: Russian Geopolitical Oil Supply Disruption Index

Note: Monthly Russian geopolitical oil supply disruption index of Iacoviello and Tong (2026), January 2000 – December 2025. Dark blue dots mark the largest spikes, selected at least four months apart, and labels report the main events in the corresponding month.
Source: Iacoviello and Tong (2026); author's selection of events.

Figure C2 plots the structural shocks identified in the QFAVAR. Shocks larger than the one used in the impulse responses are rare before 2022 and concentrate around the Russo-Georgian war of August 2008 and the annexation of Crimea in March 2014. The invasion of Ukraine produces shocks of a different order of magnitude, with the largest in March 2022, followed by large negative shocks as the index retreats from its peak.

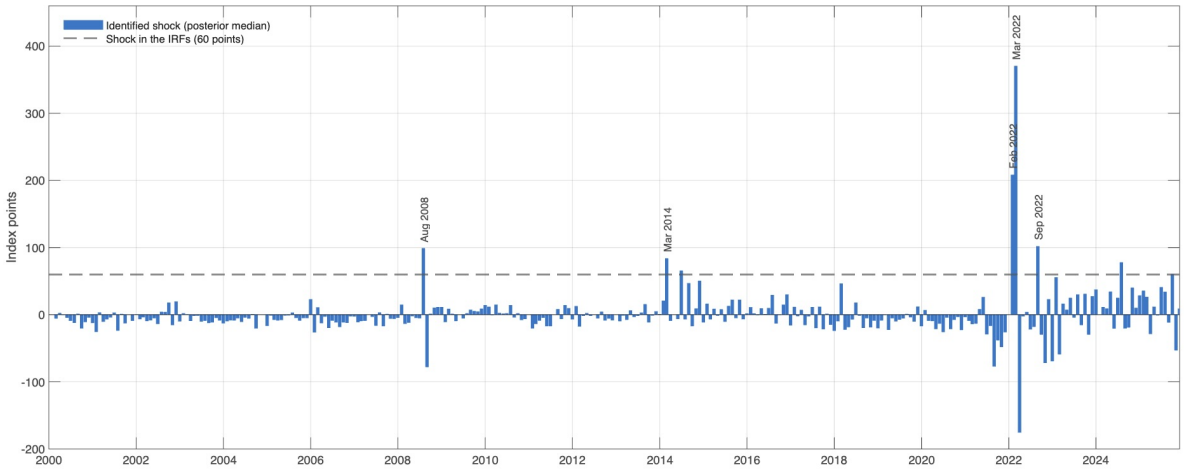


Figure C2: Identified Geopolitical Oil Supply Disruption Shocks

Note: Structural geopolitical oil supply disruption shocks identified in the QFAVAR, posterior median, in index points. The shock is the innovation to the Russia component of the geopolitical oil supply disruption subindex, identified recursively with the index ordered first. The dashed line marks the size of the shock in the impulse responses (one standard deviation of the index, about 60 points); one standard deviation of the identified shock corresponds to about 40 points. Labels mark the shocks of the three episodes considered in Section 4 (August 2008, March 2014, and March 2022) and the other largest shocks of 2022.
Sample: January 2000 – December 2025.

D Summary Statistics of Inflation Rates

Table D3: Summary Statistics of HICP Inflation

			Minimum		Maximum	
	Mean	Std. dev.	Value	Month	Value	Month
Austria	2.50	1.96	−0.42	Jul 2009	11.57	Jan 2023
Belgium	2.43	2.12	−1.73	Jul 2009	13.10	Oct 2022
Bulgaria	4.27	4.16	−2.55	May 2016	15.62	Sep 2022
Croatia	2.88	2.68	−1.46	Aug 2016	12.94	Nov 2022
Cyprus	1.98	2.30	−2.89	Aug 2020	10.56	Jul 2022
Estonia	4.27	4.36	−2.14	Nov 2009	25.16	Aug 2022
Finland	1.92	1.67	−0.68	Sep 2015	9.06	Nov 2022
France	1.88	1.45	−0.79	Jul 2009	7.29	Feb 2023
Germany	2.07	1.90	−0.69	Nov 2020	11.55	Oct 2022
Greece	2.27	2.47	−2.86	Nov 2013	12.12	Sep 2022
Ireland	2.00	2.37	−2.95	Sep 2009	9.66	Jul 2022
Italy	2.12	2.06	−1.08	Sep 2020	12.59	Oct 2022
Latvia	4.16	4.96	−4.36	Feb 2010	22.04	Sep 2022
Lithuania	3.34	4.45	−1.87	Feb 2003	22.51	Sep 2022
Luxembourg	2.48	1.86	−1.59	May 2020	10.34	Jun 2022
Malta	2.28	1.58	−1.07	Apr 2007	7.43	Oct 2022
Netherlands	2.45	2.40	−1.04	Oct 2023	17.06	Sep 2022
Portugal	2.20	2.06	−1.80	Sep 2009	10.56	Oct 2022
Slovakia	4.00	3.78	−0.86	Jul 2016	16.78	Mar 2000
Slovenia	3.31	3.03	−1.36	May 2020	11.70	Jul 2022
Spain	2.39	2.01	−1.46	Jan 2015	10.70	Jul 2022
Euro area (average)	2.72	2.30	−0.70	Jan 2015	12.44	Oct 2022

Note: Year-on-year growth rate of the headline HICP, in percent, monthly data, January 2000 – December 2025 (312 observations). Euro area (average) is the unweighted cross-country average of the 21 member states. Month reports the month in which the minimum or the maximum occurs.

Source: Eurostat; author's calculations.

E Additional Details on the Model

Units of the Impulse Responses. All variables enter the model standardized, so the impulse responses are initially expressed in standard deviations rather than in percentage points. This subsection sets out the transformation used to report them in percentage points of annual HICP inflation, and clarifies how the responses should be read.

Let $\overline{\pi}_i$ and σ_i^{π} denote the sample mean and standard deviation of inflation in country i over the estimation sample. The panel enters the measurement equation as

$$\pi_{it} = \frac{\sigma_{it}^{raw} - \overline{\pi}_i}{\sigma_i^{\pi}} \tag{E1}$$

and the global variables are standardized analogously. Because (E1) is applied country by country, it absorbs the heterogeneity in average inflation across member states, which ranges from about 1.9% in France to 4.3% in Bulgaria over the sample. This is why no intercepts appear in the measurement or state equations. They are zero by construction.

Size of the shock. The impulse responses are normalized so that the Russian geopolitical oil supply disruption index rises by one on impact in standardized units, that is, by one unconditional standard deviation of the index, $\sigma^{\mathcal{S}}$, which equals about [60] index points in the full sample. All responses are therefore read as responses to a shock of this size. Because $\sigma^{\mathcal{S}}$ is computed over the estimation sample, it differs across the samples compared in Subsection 4.1. Since the model is linear, responses are proportional to the size of the shock, so I rescale them to a common shock of c index points by multiplying them by $c/\sigma^{\mathcal{S}}$, where $\sigma^{\mathcal{S}}$ is the standard deviation of the index in the corresponding sample; in each sample, responses are converted to percentage points using the standard deviations of inflation in that sample.

From standard deviations to percentage points. The country responses $\phi_{i,h}^{(q,s)}$ are expressed in standard deviations of country i's inflation. Since (E1) is a linear transformation with fixed coefficients, responses in percentage points are recovered by reversing the scaling,

$$\phi_{i,h}^{pp,(q,s)} = \sigma_i^{\pi} \phi_{i,h}^{(q,s)} \tag{E2}$$

For the quantile factors, which are common to the panel and not attached to any single country, σ_i^{π} is replaced by the cross-sectional average $\overline{\sigma} = N^{-1} \sum_{i=1}^{N} \sigma_i^{\pi}$, so that

factor responses are expressed in percentage points for an average member state. The responses of the global variables are rescaled by their own standard deviations and are reported in their natural units.

The transformation is applied draw by draw, before the risk measures are computed. Since σ_i^{π} is a constant, it rescales the posterior distribution and leaves credible bands proportional. It changes the units of the reported responses but not their statistical significance.

Two further points are worth making explicit. First, the demeaning in (E1) plays no role. An impulse response is a difference between two conditional paths,

$$\phi_{i,h} = \mathbb{E}\left[\pi_{i,t+h}^{raw} | \varepsilon_t^{\mathcal{S}}\right] - \mathbb{E}\left[\pi_{i,t+h}^{raw}\right] \tag{E3}$$

where $\overline{\pi}_i$ enters both terms and cancels. The responses are therefore read as deviations from the path inflation would otherwise have followed, in percentage points, with no adjustment for the mean. The same holds for the risk measures, which are themselves differences of impulse responses.

Second, because σ_i^{π} differs across countries, the same standardized response translates into a larger response in percentage points for countries with more volatile inflation. The standard deviations range from 1.45 in France to 4.96 in Latvia, so the cross-country pattern in the reported responses reflects both the estimated loadings and differences in unconditional volatility. Where the object of interest is sensitivity to the shock rather than its magnitude in percentage points, the standardized responses $\phi_{i,h}^{(q,s)}$ are the appropriate comparison.

Risk measures and their responses to the shock. For a given horizon h, I summarize the conditional distribution of inflation with three measures of risk. Upside inflation risk is the distance between the right tail and the median, $\mathcal{U}(\pi_{t+h}) = \mathcal{Q}_{0.9}(\pi_{t+h}) - \mathcal{Q}_{0.5}(\pi_{t+h})$, and captures the scope for inflation outcomes well above the central tendency. Downside inflation risk is the distance between the median and the left tail, $\mathcal{D}(\pi_{t+h}) = \mathcal{Q}_{0.5}(\pi_{t+h}) - \mathcal{Q}_{0.1}(\pi_{t+h})$, and captures the scope for low-inflation outcomes. Inflation uncertainty is the total width of the conditional distribution, $\mathcal{W}(\pi_{t+h}) = \mathcal{Q}_{0.9}(\pi_{t+h}) - \mathcal{Q}_{0.1}(\pi_{t+h})$.

The model provides a natural decomposition of inflation risk through the responses of these measures to a Russian geopolitical oil supply disruption shock. Given the impulse responses of the three conditional quantiles, I define the response of each risk measure at horizon h as follows.

1. **Response of upside inflation risk.** The response of the right tail relative to the median,
$$\mathcal{U}_h = \frac{\partial \mathcal{Q}_{0.9}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}} - \frac{\partial \mathcal{Q}_{0.5}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}}$$
A positive value indicates that the shock disproportionately fattens the right tail of the inflation distribution relative to the median.
2. **Response of downside inflation risk.** The response of the median relative to the left tail,
$$\mathcal{D}_h = \frac{\partial \mathcal{Q}_{0.5}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}} - \frac{\partial \mathcal{Q}_{0.1}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}}$$
A positive value indicates that the median rises by more than the left tail, so that the lower part of the distribution stretches away from the centre and downside risk increases relative to the median; a negative value indicates that the left tail is pulled towards the median.
3. **Response of inflation uncertainty.** The response of the total width of the conditional distribution,
$$\mathcal{W}_h = \frac{\partial \mathcal{Q}_{0.9}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}} - \frac{\partial \mathcal{Q}_{0.1}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}}$$
A positive value indicates that the shock widens the conditional distribution, raising inflation uncertainty regardless of direction. By construction $\mathcal{W}_h = \mathcal{U}_h + \mathcal{D}_h$[12], so the three responses are not independent: the whole-distribution response is the sum of the two directional components.

Computation. The responses of the risk measures are computed draw by draw from the posterior distribution of the quantile factor impulse responses, after the conversion to percentage points in (E3). Let $\phi_h^{(q,s)}$ denote the posterior draw s of the IRF of the q-th quantile factor at horizon h, for $s = 1, \ldots, S$ with $S = 1000$. The posterior draws of the three risk measure responses are:
$$\widehat{\mathcal{U}}_h^{(s)} = \phi_h^{(0.9,s)} - \phi_h^{(0.5,s)}$$

[12]Sum the response of upside inflation risk and the response of downside inflation risk:
$$\begin{aligned}\mathcal{U}_h + \mathcal{D}_h &= \left(\frac{\partial \mathcal{Q}_{0.9}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}} - \frac{\partial \mathcal{Q}_{0.5}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}}\right) + \left(\frac{\partial \mathcal{Q}_{0.5}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}} - \frac{\partial \mathcal{Q}_{0.1}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}}\right) \\ &= \frac{\partial \mathcal{Q}_{0.9}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}} - \frac{\partial \mathcal{Q}_{0.1}(\pi_{t+h})}{\partial \varepsilon_t^{\mathcal{S}}} = \mathcal{W}_h\end{aligned}$$

$$\widehat{\mathcal{D}}_h^{(s)} = \phi_h^{(0.5,s)} - \phi_h^{(0.1,s)}$$
$$\widehat{\mathcal{W}}_h^{(s)} = \phi_h^{(0.9,s)} - \phi_h^{(0.1,s)}$$

Point estimates and credible bands are then obtained as:

$$\widehat{\text{Risk}}_h = \frac{1}{S}\sum_{s=1}^{S} \widehat{\text{Risk}}_h^{(s)}, \quad \text{CI}_h = \left[Q_{0.16}\left(\widehat{\text{Risk}}_h^{(s)}\right), Q_{0.84}\left(\widehat{\text{Risk}}_h^{(s)}\right) \right]$$

where $\widehat{\text{Risk}}_h^{(s)}$ stands for any of $\widehat{\mathcal{U}}_h^{(s)}$, $\widehat{\mathcal{D}}_h^{(s)}$, or $\widehat{\mathcal{W}}_h^{(s)}$, and $Q_p(\cdot)$ denotes the p-th sample quantile of the S posterior draws. This approach correctly propagates the joint posterior uncertainty of the two quantile IRFs into the credible bands of each risk measure response; computing the differences after averaging would yield the same point estimates but would understate uncertainty by ignoring the covariance between the two quantile IRFs across draws.

Country-level risk measure responses are computed analogously, replacing $\phi_h^{(q,s)}$ with the country-specific responses recovered through the measurement equation:

$$\phi_{i,h}^{(q,s)} = \gamma_{i(q)}^{(s)} \frac{\partial \Pi_{t+h,(q)}^{(s)}}{\partial \varepsilon_t^{\mathcal{S}}} + \boldsymbol{\psi}_{i(q)}^{(s)} \frac{\partial \boldsymbol{\mathcal{G}}_{t+h}^{(s)}}{\partial \varepsilon_t^{\mathcal{S}}},$$

where $\gamma_{i(q)}^{(s)}$ is the loading of country i on the q-th quantile factor $\Pi_{t,(q)}$ and $\boldsymbol{\psi}_{i(q)}^{(s)}$ is its vector of loadings on the global variables $\boldsymbol{\mathcal{G}}_{\boldsymbol{t}}$. Both are taken from the same posterior draw s, so that uncertainty in the measurement equation propagates into the country-level responses.

F Counterfactual Histories

This appendix describes how the counterfactual histories are constructed. The question is what the distribution of inflation would have looked like had the geopolitical disruption shocks to Russian oil supply been absent over a given window, with all other shocks equal to their estimated values.

Let $\boldsymbol{y_t} = [\boldsymbol{\mathcal{G}}_t', \boldsymbol{\Pi}_t']'$ denote the state vector of the VAR in (2), with reduced-form innovations $\boldsymbol{e}_t = \boldsymbol{y}_t - \sum_{j=1}^{p} \boldsymbol{A}_j \boldsymbol{y}_{t-j}$. For each posterior draw, the innovations are computed from the draw of the quantile factors $\boldsymbol{\Pi}_t$, the observed global factors $\boldsymbol{\mathcal{G}}_t$, and the draw of the autoregressive coefficients. Because the index $\mathcal{S}_t$ is ordered first and $\boldsymbol{B}_0$ is lower triangular, its reduced-form innovation is proportional to the structural shock,

$$e_t^{\mathcal{S}} = b_{11} \varepsilon_t^{\mathcal{S}}$$

where b_{11} is the first diagonal element of $\boldsymbol{B}_0$. The realized disruption shocks are therefore recovered directly from the first element of $\boldsymbol{e}_t$, without requiring the other structural shocks to be identified.

Let $\boldsymbol{\Theta}_h$ denote the response of $\boldsymbol{y}_{t+h}$ to a disruption shock that raises $\mathcal{S}_t$ by one unit on impact, $\boldsymbol{\Theta}_0 = \boldsymbol{B}_{0,\cdot 1} / b_{11}$, with $\boldsymbol{\Theta}_h = \sum_{j=1}^{\min(p,h)} \boldsymbol{A}_j \boldsymbol{\Theta}_{h-j}$ for $h \geq 1$. In a linear VAR, the contribution of the disruption shocks realized between months t_0 and t_1 to the state vector at time t is

$$\boldsymbol{\Delta}_t = \sum_{s=t_0}^{\min(t,\, t_1)} \boldsymbol{\Theta}_{t-s}\, e_s^{\mathcal{S}}, \qquad t \geq t_0,$$

and zero before t_0. The counterfactual state is $\tilde{\boldsymbol{y}}_t = \boldsymbol{y}_t - \boldsymbol{\Delta}_t$. This is equivalent to simulating the VAR forward from t_0 with the disruption shocks set to zero in the window and all other shocks kept at their estimated values; for each run, I verify numerically that the two computations coincide.[13]

The fitted and counterfactual quantiles of inflation in country i follow from the measurement equation (1),

$$\mathcal{Q}_{q,it} = \mu_i + \sigma_i \left(\gamma_{i(q)} \Pi_{t,(q)} + \boldsymbol{\psi}_{i(q)} \boldsymbol{\mathcal{G}}_t \right),$$
$$\tilde{\mathcal{Q}}_{q,it} = \mu_i + \sigma_i \left(\gamma_{i(q)} \tilde{\Pi}_{t,(q)} + \boldsymbol{\psi}_{i(q)} \tilde{\boldsymbol{\mathcal{G}}}_t \right),$$

where μ_i and σ_i are the sample mean and standard deviation of inflation in country i, which convert the standardized model back into percent. The contribution of the

[13] If the covariance matrix of the innovations varies over time, $\boldsymbol{\Theta}_0$ is computed from the impact matrix of month s for each shock $e_s^{\mathcal{S}}$.

disruption to the q-th quantile is $\mathcal{Q}_{q,it} - \tilde{\mathcal{Q}}_{q,it} = \sigma_i\left(\gamma_{i(q)}\Delta^{\Pi}_{t,(q)} + \boldsymbol{\psi}_{i(q)}\boldsymbol{\Delta}^{\mathcal{G}}_t\right)$. It combines the effect of the shocks on the common quantile factor and their effect on the global factors, weighted by the country-specific loadings. Upside and downside risk are computed from the fitted and counterfactual quantiles in the same way as in the main text. Euro Area values are cross-country averages of the member-state quantiles, with equal weights.

All quantities are computed for each posterior draw of the factors, the loadings, the autoregressive coefficients, and the innovation covariance matrix. I report posterior means and 16th–84th percentile credible bands. Averages over the window are computed draw by draw before taking posterior moments.

The baseline sets to zero all disruption shocks from February 2022 to December 2023, positive and negative. Removing only the positive realizations would yield a counterfactual without escalations, but because the shocks have mean zero, it would also attribute a positive contribution to the disruption mechanically, in any window. Removing all shocks avoids this selection. The window starts with the invasion, when the disruption to Russian oil supply ceased to be only a threat, and ends in December 2023, by which time the main sanctions and embargoes were in place and the index had returned close to its pre-war level. Disruption shocks realized before February 2022, including those associated with the build-up of Russian troops at the Ukrainian border, are kept in the counterfactual, so that the estimated contribution is, if anything, conservative. The effects of the shocks removed within the window continue to propagate after December 2023 and are included in the counterfactual path.

The counterfactual inherits three limitations. First, the model is linear, so the effects of the large shocks of 2022 are scaled up from the average effect estimated over the whole sample. Second, the coefficients are held fixed: the counterfactual abstracts from changes in policy or behavior that the absence of the disruption might have induced (Lucas, 1976). Third, the counterfactual is computed for the fitted quantiles rather than for realized inflation, so it is informative only to the extent that the model tracks the data, which it does closely over the period of interest.

G Nonlinear VARX: Impulse Responses

Let $\boldsymbol{B} = [\boldsymbol{c}, \boldsymbol{D}_1, \boldsymbol{D}_2, \boldsymbol{b}, \boldsymbol{d}]'$ collect the coefficients of the VARX, and let $\boldsymbol{X}$ denote the matrix of regressors. For each posterior draw $i = 1, \ldots, N$ of the QFAVAR, I recover the identified shock $\varepsilon_t^{\mathcal{S}}$, estimate $\boldsymbol{B}$ by OLS, and draw the coefficients from their sampling distribution,

$$\operatorname{vec}\big(\boldsymbol{B}^{(i)}\big) \sim \mathcal{N}\Big(\operatorname{vec}\big(\hat{\boldsymbol{B}}^{(i)}\big),\ \hat{\boldsymbol{\Sigma}}_u^{(i)} \otimes \big(\boldsymbol{X}^{(i)\prime}\boldsymbol{X}^{(i)}\big)^{-1}\Big),$$

where $\hat{\boldsymbol{\Sigma}}_u^{(i)}$ is the estimated covariance matrix of the VARX residuals. Credible bands therefore reflect both the uncertainty about the identified shock and the factors, and the sampling uncertainty of the second-step estimates.

Shock sizes are set in index points. A shock of P points corresponds to

$$\delta^{(i)} = \frac{P}{\sigma_{\mathcal{S}}\, s_{\varepsilon}^{(i)}}$$

units of $\varepsilon_t^{\mathcal{S}}$, where $\sigma_{\mathcal{S}}$ is the sample standard deviation of the index and $s_{\varepsilon}^{(i)}$ is the standard deviation of the innovation to the standardized index in draw i. Since the shock enters the VARX only contemporaneously, the response of $\boldsymbol{z}_{t+h}$ is

$$\boldsymbol{r}_0(\delta) = \boldsymbol{b}\,\delta + \boldsymbol{d}\,\delta^2, \qquad \boldsymbol{r}_h(\delta) = \sum_{j=1}^{\min(2,h)} \boldsymbol{D}_j\, \boldsymbol{r}_{h-j}(\delta), \quad h = 1, \ldots, H,$$

in standardized units. Responses are converted into the units of each variable with $\boldsymbol{\Lambda} = \operatorname{diag}(\sigma_{oil}, \sigma_{gas}, \sigma_{\mathcal{F}}, \sigma_{gscpi}, \sigma_u, \sigma_R, \bar{\sigma}_\pi, \bar{\sigma}_\pi, \bar{\sigma}_\pi)$, where the first six elements are the standard deviations of the global factors and $\bar{\sigma}_\pi$ is the cross-country average standard deviation of inflation. To compare shocks of different size, the responses in Figures 13 and 14 are expressed per 100 index points,

$$\tilde{\boldsymbol{r}}_h(P) = \frac{100}{P}\,\boldsymbol{\Lambda}\,\boldsymbol{r}_h\big(\delta(P)\big),$$

so that in a linear model ($\boldsymbol{d} = \boldsymbol{0}$) the normalized responses to small and large shocks coincide. The responses of upside and downside risk are the differences between the normalized responses of the 90th percentile and the median, and of the median and the 10th percentile, computed draw by draw.

H Additional Figures

H.1 Responses of Global Factors to a Geopolitical Disruption to Russian Oil Supply

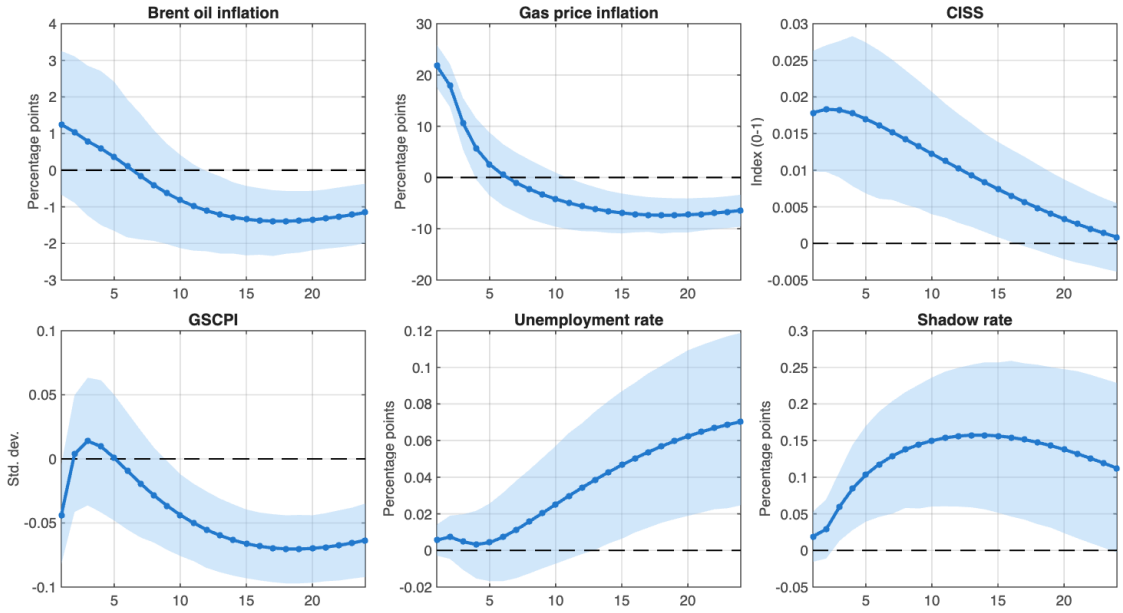


Figure H3: Dynamic Responses of Global Factors to a Geopolitical Disruption to Russian Oil Supply

Note: Impulse responses of the global factors to a geopolitical disruption to Russian oil supply, estimated with the QFAVAR. The shock is identified recursively, with the Russia component of the geopolitical oil supply disruption subindex of Iacoviello and Tong (2026) ordered first. Responses are scaled to a shock that raises the index by one standard deviation on impact and are expressed in the units of each variable. Brent oil and European gas prices enter as year-on-year growth rates of prices in euros, so their responses are in percentage points. The CISS is an index bounded between 0 and 1. The GSCPI is measured in standard deviations from its average value. The unemployment rate and the shadow rate are in percentage points. Solid lines denote posterior means; shaded areas denote the 16th–84th percentile posterior credible bands.
Sample: January 2000 – December 2025.

Figure H3 reports the responses of the global factors in the QFAVAR, which underlie the discussion of the supply and demand channels in Section 3 and the signs in Table 1. The index itself rises by one standard deviation, about 60 points, on impact and returns to its pre-shock level within about ten months. Gas price inflation rises by about 20 percentage points on impact and returns to zero after about five months,

whereas oil price inflation does not respond significantly on impact. Both turn negative at longer horizons, reflecting in part the year-on-year measurement. Financial stress rises on impact and remains elevated for more than a year, supply chain pressures ease, and unemployment increases gradually, becoming significant after about a year. The shadow rate rises significantly after a few months and peaks at about 0.15 percentage points after about a year.

H.2 Subsample Analysis

This appendix reports additional results for the samples ending in December 2019 and December 2021 discussed in Subsection 4.1. In all samples, the shock raises the index by 60 points on impact.

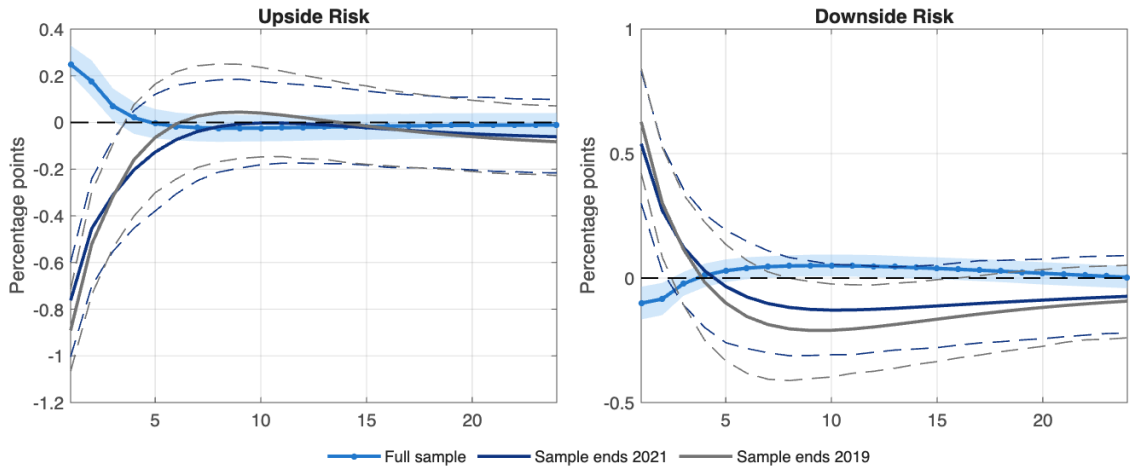


Figure H4: Responses of Euro Area Inflation Tail Risks Across Estimation Samples

Note: Impulse responses of upside risk (90th percentile minus median) and downside risk (median minus 10th percentile) of Euro Area inflation to a geopolitical shock to Russian oil supply, estimated with the QFAVAR on the full sample (January 2000 – December 2025; blue, shaded band) and on samples ending in December 2021 (red) and December 2019 (green; dashed bands). The shock is identified recursively, with the Russian geopolitical oil supply disruption index of Iacoviello and Tong (2026) ordered first, and raises the index by 60 points on impact in all samples. Responses are expressed in percentage points, using the cross-country average standard deviation of inflation in each sample. Solid lines denote posterior means; bands denote the 16th–84th percentile posterior credible bands.

Figure H4 reports the responses of upside and downside risk. In the pre-war samples, neither measure responds significantly at any horizon, and the credible bands are wide on impact. As discussed in the main text, in these samples the data identify the

joint response of the median and the upper quantile factors but not their difference, so the impact responses of the risk measures are imprecisely estimated.

Figure H5 reports the responses of the global factors, which help interpret the contrast between samples. Before 2022, the shock has the signature of a demand shock. Brent prices fall by about 10 percent after five to six months, European gas prices decline with a delay, financial stress rises temporarily, supply chain pressures ease, and the shadow rate falls by about 0.35 percentage points after eight to nine months, indicating monetary easing. In the full sample, by contrast, gas prices rise on impact and the shadow rate does not respond, consistent with the dominance of the supply channel. The different responses of energy prices and monetary policy thus account for the opposite sign of the inflation response across samples.

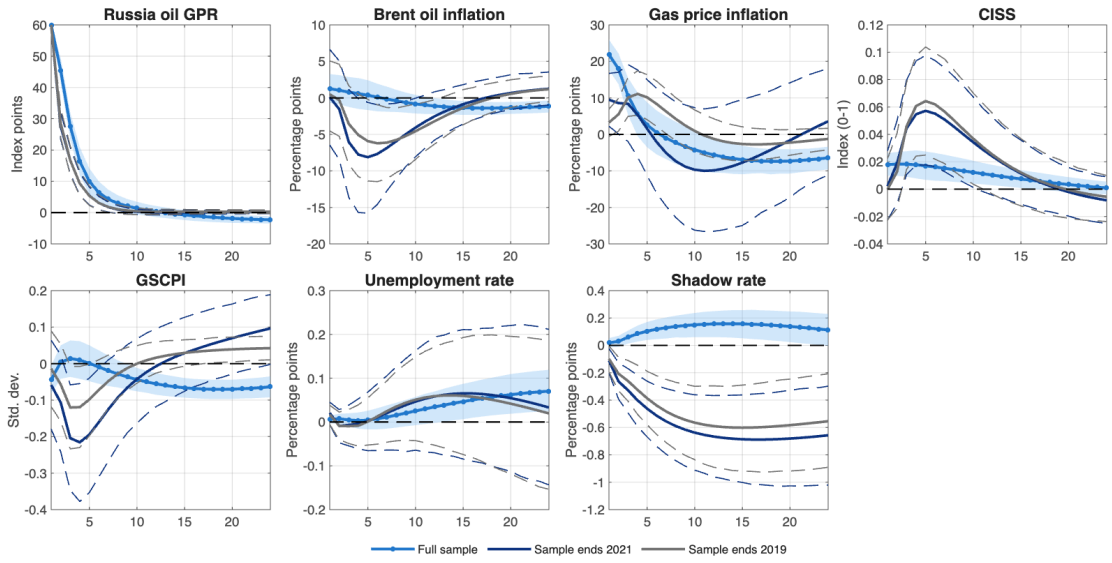


Figure H5: Responses of the Global Factors Across Estimation Samples

Note: Impulse responses of the global factors to a geopolitical shock to Russian oil supply, estimated with the QFAVAR on the full sample (January 2000 – December 2025; blue, shaded band) and on samples ending in December 2021 (red) and December 2019 (green; dashed bands). The shock raises the index by 60 points on impact in all samples. Brent oil and European gas prices enter as 100 times their logarithm, so their responses are in percent. The CISS is an index bounded between 0 and 1. The GSCPI is measured in standard deviations from its average value. The unemployment rate and the shadow rate are in percentage points. Solid lines denote posterior means; bands denote the 16th–84th percentile posterior credible bands.

H.3 Contribution of Geopolitical Disruptions to Euro Area Inflation

Figure H6 reports the contribution of the geopolitical disruption shocks of February 2022 – December 2023 to the 10th, 50th, and 90th percentiles of euro area inflation, computed as the difference between the fitted percentiles and their counterfactual paths in Figure 11. The contribution to the 90th percentile appears immediately, reaching about 1.8 percentage points in March 2022, while the 10th percentile initially falls slightly, so that the disruption first widens the upper part of the distribution. The contributions to the median and the 10th percentile build up over the following months and peak at about two percentage points in mid-2022. From mid-2023, the contribution turns negative at all three percentiles, reaching about -0.5 to -0.75 percentage points in 2024 and 2025, reflecting the reversal of energy prices and the tighter monetary policy that followed the disruption.

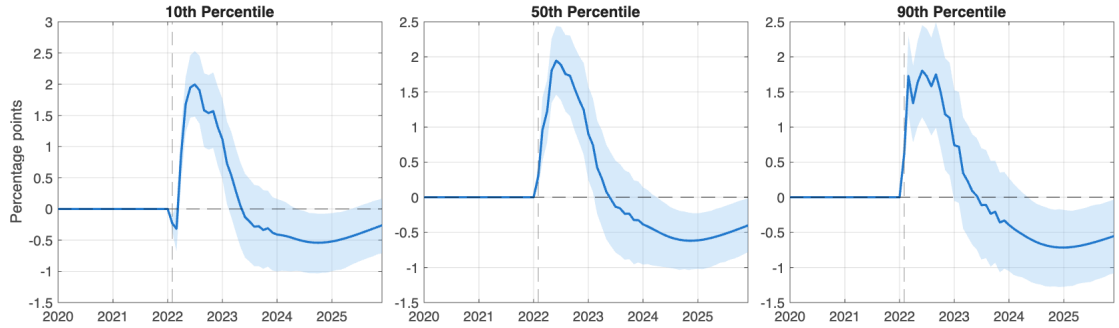


Figure H6: Contribution of Geopolitical Disruptions to Euro Area Inflation Percentiles

Note: Difference between the fitted 10th, 50th, and 90th percentiles of euro area inflation and their counterfactual paths without the geopolitical disruption shocks of February 2022 – December 2023, in percentage points. Euro area percentiles are the unweighted cross-country averages of the member-state percentiles. The solid line denotes the posterior mean; the shaded area denotes the 16th–84th percentile credible band. The dotted vertical line marks February 2022.
Sample: January 2000 – December 2025.

H.4 Contributions by Member State

Figure H7 reports, for each member state, the monthly contribution of the geopolitical disruption shocks of February 2022 – December 2023 to the 10th, 50th, and 90th percentiles of inflation. The timing of the contributions mirrors that of the impulse re-

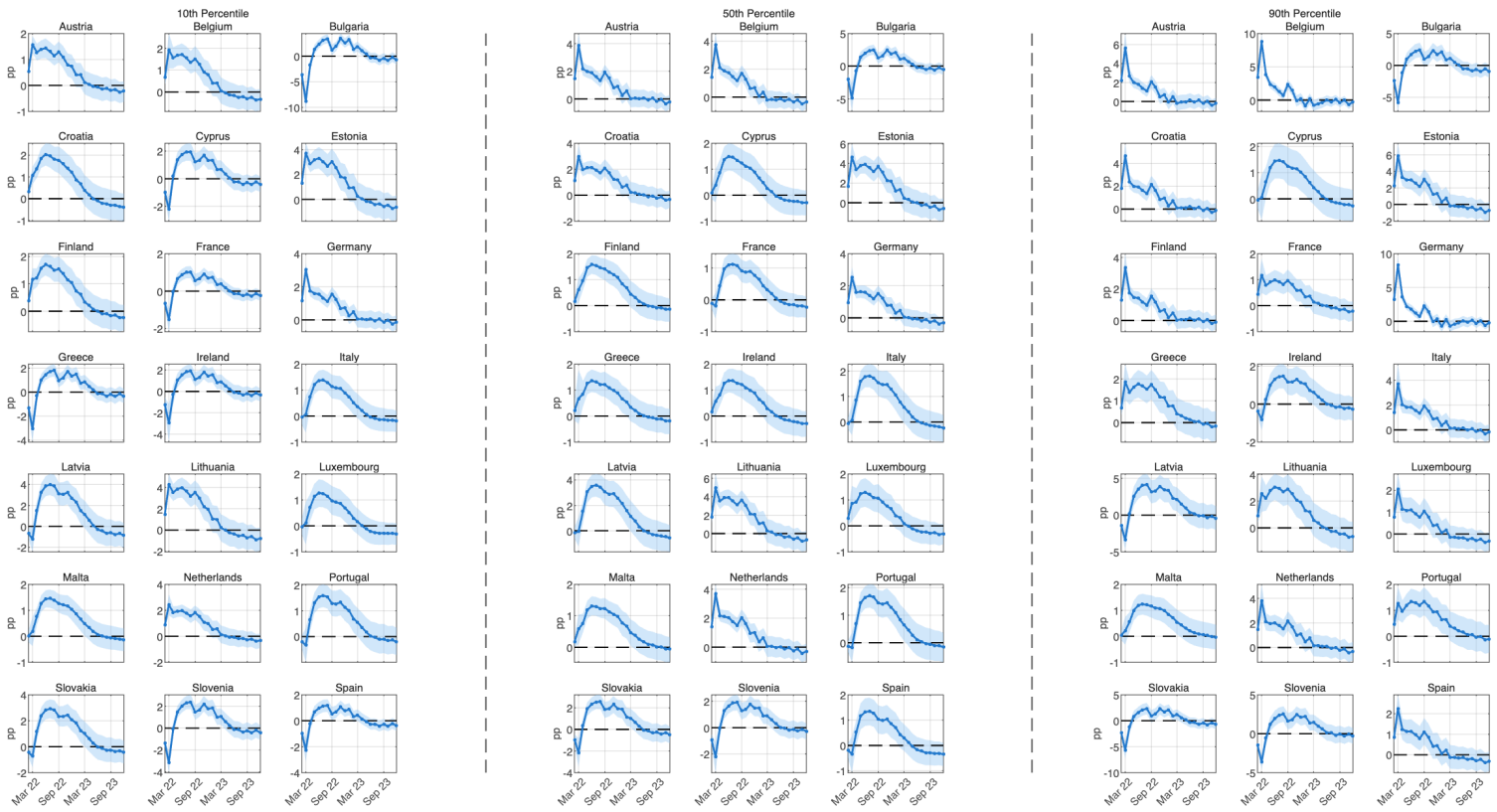


Figure H7: Monthly Contribution of the Geopolitical Disruption to Inflation by Member State

Note: Contribution of the geopolitical disruption shocks of February 2022 – December 2023 to the 10th, 50th, and 90th percentiles of inflation in each member state, computed as the difference between the fitted percentile and its counterfactual path, month by month, in percentage points. Solid lines denote posterior means; shaded areas denote 16th–84th percentile posterior credible bands. Note the differing vertical scales across panels.

Sample: January 2000 – December 2025.

sponses in Section 3. In Germany, Belgium, and Austria, the contribution to the 90th percentile peaks in March 2022, immediately after the invasion, at about 5.7 to 8.8 percentage points, and declines steadily thereafter. In the member states in which the response builds up gradually, including Bulgaria, Slovakia, Slovenia, and Greece, the contribution is negative at the onset of the war, most markedly at the 10th percentile, turns positive within two to three months, and peaks in the summer of 2022. In almost all member states, the contributions fade by the end of 2023 and turn slightly negative, consistent with the counterfactual path lying above the fitted one in 2024 and 2025.

H.5 Responses to Disruptions of Historical Size

Figure H8 reports the total responses, estimated with the nonlinear VARX of Section 4.3, to shocks equal to those identified at the time of the main historical episodes. The shocks identified in August 2008, March 2014, and March 2022 amount to about 99, 84, and 371 index points, close to the monthly increases in the index in those months, against about 40 points for a one-standard-deviation shock. The disruption of 2022 was therefore about nine times as large as a typical shock, but its effects on the inflation percentiles are between 20 and 40 times as large. The responses also differ in kind. After shocks of the size observed before the war, supply chain pressures ease and oil price inflation barely moves, whereas after the 2022 disruption oil price inflation rises sharply and supply chain pressures increase for several months, consistent with the supply channel prevailing only for large disruptions.

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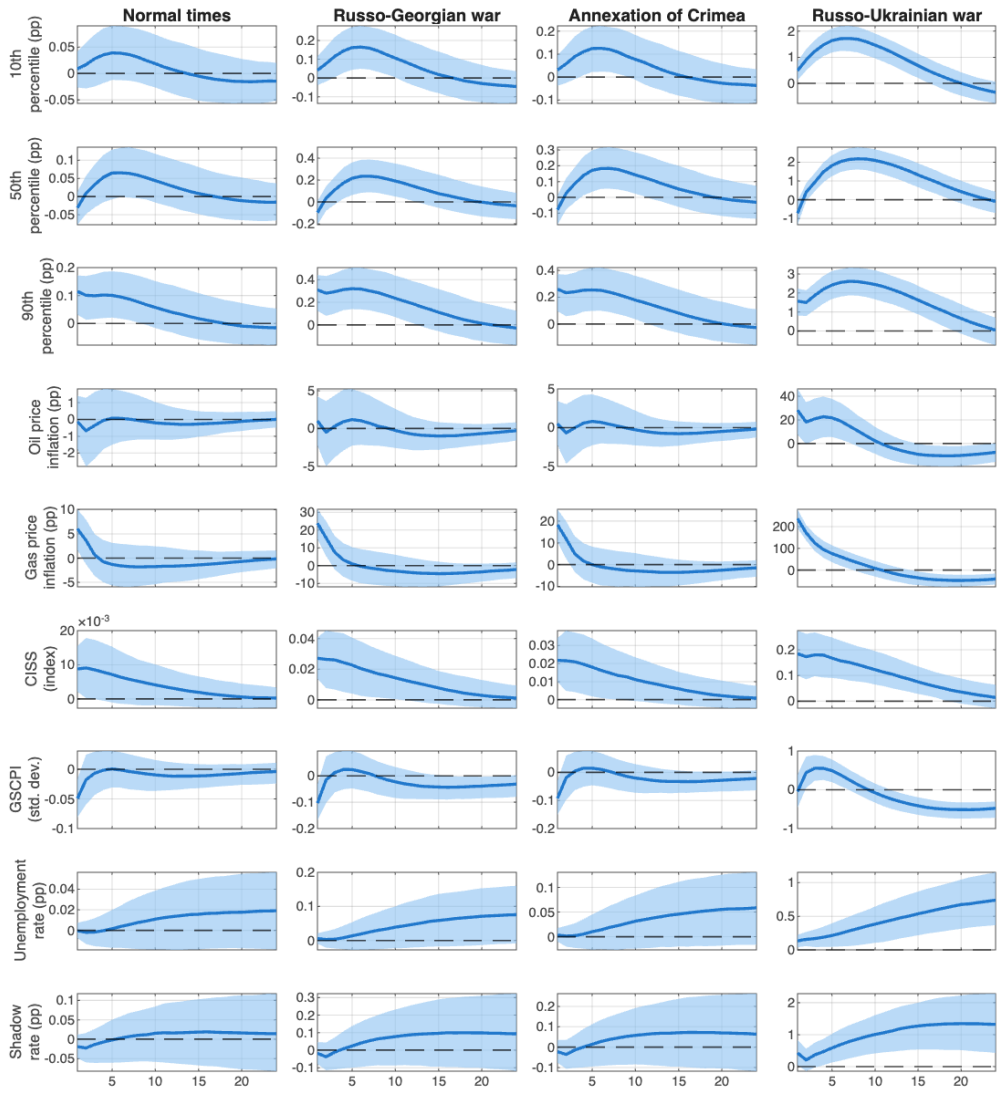


Figure H8: Responses to Geopolitical Disruptions of Historical Size

Note: Total responses, estimated with the nonlinear VARX, of the 10th, 50th, and 90th percentile factors of euro area inflation and of the global factors to a geopolitical disruption to Russian oil supply. Columns report the responses to a one-standard-deviation shock (normal times, about 40 index points) and to shocks equal to those identified in August 2008 (Russo-Georgian war, about 99 points), March 2014 (annexation of Crimea, about 84 points), and March 2022 (Russo-Ukrainian war, about 371 points). Solid lines denote posterior medians; shaded areas denote the 16th–84th percentile credible bands across QFAVAR posterior draws. Each panel has its own vertical scale.